When Is Surrogate-Assisted Multiobjective Ensemble Weighting Trustworthy? A Replicated Study of Mixture Designs, Scheffé Surfaces and Normal Boundary Intersection on the Classifier-Weight Simplex*

Abstract

Choosing soft-voting ensemble weights is a mixture experiment already addressed by an established mixture-design, response-surface and Normal Boundary Intersection pipeline. The artificial-intelligence contribution is a diagnosis of when that pipeline can be trusted, not a new method; the engineering application is designing weighted classifier ensembles under a measured deployment cost. Three arms differing in objective and anchor sources ran on 30 overlapping, paired partitions of four public financial-services datasets, with controls the framework lacks: a reliability gate fixed beforehand, real-objective revalidation, a reference whose sampled core contains no output of the compared methods, and an untouched holdout. Gate passes ranged from 0 to 30 of 30 partitions by dataset and metric. Under the continuous cost the arms optimize, real anchors beat surrogate ones on both primary endpoints in every partition of the two datasets whose surfaces for the area under the receiver operating characteristic curve almost never passed, and in 23–24 of 30 elsewhere (significant after correction only on Santander). On those two datasets anchor misplacement was the largest observed failure; a control traced most of it to injected extreme solutions. The gate detected unusable surfaces but not misplaced anchors. Continuous and step deployment costs disagreed on the winning candidate set in 8–24 of 30 partitions and on the anchor effect's direction on one dataset. The non-dominated sorting genetic algorithm II (NSGA-II), given about twice the metamodel-free arm's weight-vector evaluations, outperformed it. The diagnosis concerns one-shot pipelines without infill; four datasets establish a pattern, not a general rule.

Keywords: ensemble weighting, mixture design of experiments, response surface methodology, normal boundary intersection, surrogate-assisted multiobjective optimization, inference cost

1. Introduction

A weighted average of several classifiers' predicted probabilities is among the oldest and most reliable devices in applied machine learning (Hansen and Salamon, 1990; Wolpert, 1992; Kittler et al., 1998; Caruana et al., 2004). Its weights are not free parameters: to keep the output a probability they must be non-negative and sum to one, and non-negativity with normalization is what makes stacked combinations stable (Breiman, 1996). The feasible set is the unit simplex, so ensemble weighting is a *compositional* optimization problem.

That has a methodological consequence: optimization over a simplex is the subject of mixture experiments, a mature branch of experimental design (Scheffé, 1958, 1963; Cornell, 2002). The classifiers are the components, a pure vertex is a single model, the centroid uniform soft voting, and a Scheffé polynomial the natural response surface. The correspondence is exact and already exploited for portfolio weights (Mendes et al., 2016; Leal et al., 2022) and neural-network ensembles (Moreira et al., 2021), and combined with Normal Boundary Intersection for

*Abbreviations: CHIM, convex hull of individual minima; DoE, design of experiments; NBI, Normal Boundary Intersection; NSGA-II, non-dominated sorting genetic algorithm II; ROC-AUC, area under the receiver operating characteristic curve; RSM, response surface methodology.

forecast combination (Bacci et al., 2019), load forecasting (Rocha et al., 2025) and multiobjective engineering optimization (Pereira et al., 2025). Most directly, Kwon et al. (2024) treat five heterogeneous *classifiers*' weights as mixture components and maximize a Scheffé polynomial of accuracy or F_1 on eleven tabular binary datasets.

We therefore claim no part of that construction: mixture design over combination weights, a Scheffé surrogate over the simplex, and NBI on that surrogate are established prior work, individually and for classifier ensembles. What is not established is whether the construction stays *trustworthy* once the problem becomes genuinely multiobjective. That is the question this paper addresses.

1.1. Why the transfer is not routine

The applications in which this framework matured share three properties that classifier ensembles do not.

First, *the responses are smooth*: tool life or process cost varies smoothly with cutting speed, where a low-order polynomial is a defensible local model. The area under the receiver operating characteristic curve (ROC-AUC) is instead a rank statistic, piecewise constant in the weights. A richer design alone cannot remove the approximation bias of an unsuitable model class, so whether a low-order polynomial represents such a response adequately has to be measured on unseen compositions rather than assumed.

Second, *the decision variables are the mixture.* When the simplex holds scalarization weights, as in the lineage above, NBI's anchors are optima of the surrogate in another space; when it holds the decision variables themselves, the anchors are feasible-region points (often vertices), so an anchor error displaces the geometry in which every subproblem is posed. NBI derives its whole search geometry from the payoff matrix of individual optima (Das and Dennis, 1998), so errors there propagate globally. That misplaced anchors damage the construction is established: the worst criterion values read off a payoff table can differ substantially from the true worst values over the efficient set (Isermann and Steuer, 1988), the nadir point is hard to estimate (Deb et al., 2010), and the choice of the individual minima has itself been treated as a design lever (Herrmann-Wicklmayr and Flaßkamp, 2026). What that literature, and the normalization remedies built on it (Messac et al., 2003), asks about is estimating the *true* objectives; none of it is about a surrogate standing in for them.

Third, *one objective is a step function*: a model with any non-negligible weight must be loaded and executed, one with none costs nothing, so deployment cost is not proportional to a weight. A method requiring differentiable objectives can only optimize a continuous relaxation, with no guarantee that it and the quantity actually paid agree about which solution is best.

Each can break a different part of the pipeline (the model class, the search geometry, the objective definition), and none is visible in the aggregate fit statistics such studies conventionally report.

1.2. Approach

We decompose the framework. Because the ensemble output is linear in the weights and base-model probabilities can be cached, evaluating a weight vector on the true objectives costs one matrix–vector product (0.4 to 2.1 milliseconds here), cheap enough to run the *same* NBI three times per replication, changing only where objectives and anchors come from.

- **NBI-A** optimizes the Scheffé surfaces with anchors from their own optima: the inherited construction and our control.
- **NBI-B** optimizes the same surfaces with anchors from the real single-objective optima; A versus B isolates *anchor misplacement.*

- **NBI-C** drops the metamodel and runs the same NBI on the exact out-of-fold objectives: a metamodel-free NBI in the published sense of Gellerich and Majschak (2023), whose subproblems are physical experiments; B versus C isolates residual *interior surrogate error*.

All three arms fit their surfaces once from a fixed design and then optimize. This is the *sequential* form of surrogate-assisted optimization (Tabatabaei et al., 2015): one-shot, with no infill. It is the form the lineage above uses; nothing here tests model-managed, infill-based surrogate-assisted optimizers such as ParEGO, MOEA/D-EGO or SMS-EGO (Knowles, 2006; Zhang et al., 2010; Ponweiser et al., 2008), which evaluate new points on the true objectives inside the search loop and exist precisely to repair the kind of surrogate error diagnosed here. Every conclusion in this paper is scoped to the one-shot, no-infill case (Section 2.5).

We also run NSGA-II on the same objectives, with about twice as many weight-vector evaluations as NBI-C. Four controls the lineage does not contain surround the decomposition: external validation of every surface on 100 unseen compositions, with a reliability gate fixed before the replicated benchmark; revalidation of every candidate on the true objectives before any indicator; an empirical Pareto reference whose sampled core of at least 10^5 points per replication contains no output of the compared methods and is checked for convergence, then augmented with every candidate set for scoring; and an untouched holdout never used for fitting, weighting, thresholding or selection.

Four public binary tabular datasets, namely customer-transaction prediction (Santander), insurance-claim processing (BNP Paribas Cardif), auto-insurance claim prediction (Porto Seguro) and consumer-credit default (UCI credit), are partitioned 30 times each, giving 120 replications with recorded seeds. The four prediction tasks differ, but all four come from financial-services applications, so any cross-dataset statement is made within a single application domain. Because partitions of one dataset overlap heavily, we treat the replication distribution as partition sensitivity, not a sample from a population: comparisons are paired by partition and tested with the Nadeau–Bengio correction (Nadeau and Bengio, 2003), which the descriptive bootstrap and Wilson intervals do not carry; the dataset is the unit of generalization; and nothing is pooled across the 120 replications.

1.3. Research questions

RQ1 How accurately do Scheffé surfaces predict ensemble performance on compositions they have not seen, and does this depend on the dataset, on the metric, or on both?

RQ2 How stable are the fitted surfaces across resampling, and does the classical mixture-design reading of the interaction coefficients survive confrontation with the real objectives?

RQ3 What goes wrong when NBI is anchored on surrogate optima?

RQ4 Does replacing surrogate anchors with real ones repair the front, and does the reliability gate predict when this is necessary?

RQ5 How does surrogate-assisted NBI compare with metamodel-free NBI, what does the difference cost, and how do both compare with a canonical evolutionary optimizer on the same objectives?

RQ6 How large is the actual conflict between discrimination and calibration?

RQ7 Does the definition of deployment cost change which method wins?

RQ8 Do solutions selected on out-of-fold data transfer to an untouched holdout?

RQ9 Would the first ten replications have produced the same conclusions?

1.4. Contributions

1. **A controlled decomposition of a one-shot surrogate-assisted multiobjective pipeline into surrogate error, anchor misplacement and solver behaviour**, through three NBI variants defined by two methodological factors. The ingredients are not new and are not claimed: anchor and ideal/nadir misestimation is an established

failure mode (Isermann and Steuer, 1988; Deb et al., 2010; He et al., 2021), the choice of individual minima has already been treated as a design lever (Herrmann-Wicklmayr and Flaßkamp, 2026), and NBI run metamodel-free on measured objectives with real anchors is published (Gellerich and Majschak, 2023). To the best of our literature search, no prior study computes the anchors of a method built on the convex hull of individual minima (CHIM) both from a fitted surrogate and from the true objectives on the same problem and compares the resulting fronts, so what is new here is the surrogate *provenance* of the anchor error and the controlled comparison of the two constructions; the nearest ensemble-weighting precedent (Kwon et al., 2024) optimizes one metric at a time and has no anchors or front.

2. **Evidence that, on two of the four datasets, anchor misplacement has the largest observed effect among failures attributable to the surrogate pipeline.** Under the continuous cost the arms optimize, real anchors improve both primary endpoints in 30 of 30 partitions on the two datasets whose ROC-AUC surfaces almost never pass the gate (Santander and BNP Paribas), in 23 (IGD^+) and 24 (hypervolume) of 30 on Porto Seguro, and in 24 of 30 on UCI credit; after correction the improvement is significant only on Santander. Removing the surrogate entirely, once anchors are correct, adds much less on Santander and BNP Paribas, about as much on Porto Seguro, and three times as much on UCI credit, where subproblem convergence rather than surrogate error dominates; the tie and the reversal prevent a universal claim.
3. **A demonstration that an external reliability gate detects unusable surfaces but not misplaced anchors.** On one dataset the two properties diverge: the partitions where surrogate-anchored NBI collapses are precisely those where the parsimony rule selects a ROC-AUC surface that fits held-out compositions *better* and passes the gate more often. Surface quality and argmax quality are different properties, and only the second governs NBI.
4. **A replicated evaluation architecture for ensemble-weight optimization**: external surrogate validation, real-objective revalidation, a reference whose sampled core contains no output of the compared methods, and holdout confirmation, over 4 datasets $\times$ 30 partitions with paired inference and Nadeau–Bengio-corrected primary tests. Three findings it overturned, one of them an artifact noted but not resolved in the ten-partition analysis, are reported explicitly.
5. **Two findings about how these surfaces should be read.** The classical synergism criterion $\beta_{ij} > |\beta_i - \beta_j|$ holds for the leading pair's fitted surfaces on all four datasets in every partition, yet the real 50/50 blend beats its better member on only one. And the continuous cost these methods optimize disagrees with the step cost deployment pays often enough to change the winning method in up to 24 of 30 partitions. That a change of cost accounting can reorder a comparison is known across ensemble and cascade architectures (Wang et al., 2022); what is reported here is the disagreement between two accountings applied to one and the same weight vector.

We claim neither that NBI outperforms other multiobjective optimizers (NSGA-II, given about twice the weight-vector evaluations of NBI's metamodel-free arm, outperforms that arm on all four datasets), nor that it produces more uniformly spaced fronts, nor that any method here is a general-purpose recommendation. The contribution is the set of conditions under which an established framework can and cannot be trusted in a new problem class, and the controls needed to tell the difference. Both are stated for the one-shot, no-infill form of that framework, since no arm here performs iterative infill.

2. Related work

This section keeps only what bears on a novelty claim.

2.1. Classifier ensemble weighting

Weighted voting and weighted probability averaging are long established (Hansen and Salamon, 1990; Kittler et al., 1998; Dietterich, 2000; Kuncheva, 2014); the weights come from stacked generalization (Wolpert, 1992; Breiman, 1996; LeBlanc and Tibshirani, 1996; van der Laan et al., 2007; LeDell et al., 2016), greedy forward selection (Caruana et al., 2004), numerical optimization (Zhang and Zhou, 2011; Yao et al., 2018; Shahhosseini et al., 2022; Hassan et al., 2026; Xu et al., 2026) or theory (Fumera and Roli, 2005). Throughout, the simplex is almost always respected as a constraint rather than designed over as an experimental region, and essentially all of this work is single-objective.

2.2. Multiobjective ensemble optimization

Multiobjective ensemble learning divides by decision variable: the *members* (Abbass, 2003; Chandra and Yao, 2006; Jin and Sendhoff, 2008); a *subset* (Dos Santos et al., 2008; Qian et al., 2015); or (the family relevant here) *continuous combination weights* under conflicting criteria, optimized by metaheuristics (Ekbal and Saha, 2011, 2013; Bandyopadhyay et al., 2008; Onan et al., 2016; Rezaei et al., 2019; Zhao et al., 2018; Vrugt et al., 2006; Galván et al., 2026) and surveyed in Ribeiro and Reynoso-Meza (2020).

This family is therefore *not* new, and we do not claim it. Absent from this cluster is any designed experiment and fitted response surface over the weight simplex, any classical scalarization method such as NBI, and, with the partial exception of the sparsity term in Zhao et al. (2018), any treatment of deployment cost as an objective. Fronts are rarely replicated over resampled partitions with corrected inference (Demšar, 2006).

2.3. Mixture experiments and response surfaces

Mixture experiments address exactly the constraint set of Equation (2), through simplex-lattice designs and Scheffé's canonical polynomials (Scheffé, 1958, 1963; Cornell, 2002; Myers et al., 2016); because $\sum_i w_i = 1$, their coefficients are not component effects in the ordinary sense (Piepel, 1982).

The classical reading of a positive interaction coefficient is *synergistic blending* (Cornell, 2002): a statement relative to the chord, not relative to the better pure component, and the distinction turns out to matter (Section 5.2).

Mixture designs are beginning to appear in machine learning (Karl et al., 2023; Leal et al., 2025; Xie et al., 2023; Ye et al., 2025; Liu et al., 2025; Chen et al., 2025; Thudi et al., 2025), but share our formalism and not our object: none models the performance of a weighted *prediction* ensemble.

The exception, and the closest prior art to this study, is Kwon et al. (2024): the weights of five heterogeneous classifiers as mixture components, a Scheffé canonical polynomial of accuracy or F_1 fitted over a mixture design, reduced by backward elimination and maximized subject to $\sum_i w_i = 1$, on eleven tabular binary datasets. The simplex formulation, the mixture design over classifier weights and the Scheffé surrogate are therefore established prior art for classifiers, and we claim none of them. What that study does not do is what this paper is about: it optimizes one scalar metric at a time, so it has no Pareto front, no anchors and no NBI; it validates the fitted polynomial only in-sample, against no separately sampled reference front; and it carries no cost objective.

2.4. The DoE–RSM–NBI framework and its application to ensemble weights

Normal Boundary Intersection (Das and Dennis, 1998) was introduced to obtain a distributed representation of the Pareto boundary, by solving for each convex weight vector a subproblem that advances from the convex hull of individual minima along a quasi-normal direction. It is widely combined with response-surface metamodels in engineering optimization, and a substantial body of work applies NBI to polynomial surfaces fitted from designed experiments (Brito

et al., 2014; Costa et al., 2016; Lopes et al., 2016; Naves et al., 2017; Belinato et al., 2019; de Almeida et al., 2022).

Within that tradition, mixture designs have been used to schedule the NBI weights and to model post-Pareto criteria over the weight simplex. Mendes et al. (2016) and Leal et al. (2022) model portfolio proportions with mixture designs; Bacci et al. (2019) builds a simplex lattice over time-series forecast-combination weights, fits Scheffé models to factor scores of residual accuracy metrics and applies NBI to those models; de Paula et al. (2019) crosses a mixture design over objective weights with genetic-algorithm hyperparameters. Pereira et al. (2025) extends the framework with a mixture-design post-Pareto stage in which Scheffé polynomials of a generalized distance and of the entropy of the weights are themselves optimized by a second NBI. Its anchors, like those of the rest of this literature, are the individual optima of the surrogate surfaces: the configuration we label NBI-A. A master's dissertation (Ribeiro, 2026) applies the same framework to hyperparameter optimization.

Most directly, Moreira et al. (2021) determine neural-network ensemble weights by a mixture design, and Rocha et al. (2025) combine a $\{3, 5\}$ simplex-lattice design over the mixing weights of three neural networks, a factor-analytic surrogate of correlated error metrics, NBI over those surrogates, and an entropy-based post-Pareto choice, for short-term load forecasting. That is the present pipeline applied to a regression ensemble, and its stated first contribution is casting ensemble-weight definition as a structured mixture-design problem.

We therefore claim no part of this construction as new. Our contribution begins where this literature stops: none of these studies validates its mixture surrogate on unseen compositions, computes anchors from anything other than the surrogate, revalidates its Pareto candidates on the true objectives, scores them against a separately sampled reference, or replicates over resampled partitions.

2.5. Surrogate-assisted multiobjective optimization

The broader surrogate-assisted literature (Jin, 2011; Tabatabaei et al., 2015; Chugh et al., 2019; Deb et al., 2019) answers the risk that a surrogate misleads with *model management*, and that answer is *iterative*: ParEGO, MOEA/D-EGO, SMS-EGO and K-RVEA (Knowles, 2006; Zhang et al., 2010; Ponweiser et al., 2008; Chugh et al., 2018) refit on true-objective infill inside the search loop, so a false surrogate optimum is corrected later. The pipeline studied here is that taxonomy's other member: the *sequential* framework (fit the surrogate once from a fixed design, then optimize it, with no infill; Tabatabaei et al., 2015) identified there as most exposed to surrogate error. That is what the lineage of Section 2.4 does and what we replicate, so our diagnosis is a diagnosis of *one-shot, no-infill* surrogate pipelines. It is not a test of model-managed, infill-based surrogate-assisted optimizers and must not be read as one: by the standards of that literature the exposure of the surrogate-anchored arm is anticipated rather than surprising, and our real-anchor arm is best understood as a single targeted correction (one model-management step applied to the anchors alone) rather than as adaptive infill.

Extreme-point reliability has been studied, but on the other side of the surrogate divide. Isermann and Steuer (1988) showed computationally that the worst criterion values in a payoff table built from individual optima can differ substantially from the true worst values over the efficient set; nadir and front-range mis-estimation degrades performance across MOEAs (Deb et al., 2010; He et al., 2021); and Herrmann-Wicklmayr and Flaßkamp (2026) treat the *choice* of individual minima as a design decision, replacing extreme with non-extreme minima to refine the utopia–nadir hyperbox. That misplaced anchors damage a scalarization is therefore established, and we claim neither the mechanism nor its direction as a discovery. In all of that work the extreme points are computed or estimated on the *true* objectives; none of it concerns anchor error introduced by a metamodel. The normalized normal constraint method (Messac et al., 2003) derives its normalization and its constraints from the individual minima, as NBI does, so on fitted surfaces it too would take its anchors from the surrogate; our protocol was frozen with NBI as the front generator, so we do not evaluate whether the effect we report carries over to it.

To the best of our literature search, no prior study computes a CHIM method's anchors both from a fitted surrogate and from the true objectives on the same problem and compares the resulting fronts; the gap concerns the surrogate *provenance* of the anchor error and that controlled contrast, not whether anchors matter. The closest external analogue we located is the NBI-A pattern itself: Abolghasemian et al. (2022) runs a modified NBI on regression metamodels validated by fit statistics alone, and reports no re-evaluation of its Pareto solutions on the simulation. At the other extreme, Gellerich and Majschak (2023) derive the experimental plan from the NBI subproblems themselves and evaluate each subproblem by physical test, with no metamodel anywhere and anchors that are real experimental optima. A metamodel-free NBI is thus published prior art and the idea of our NBI-C arm is not claimed as new; because that study has no surrogate counterpart on its problem, it makes no comparison. The two papers bracket our NBI-A and NBI-C arms without either of them running the contrast.

2.6. Cost-aware ensemble deployment

Prediction-time cost has been an explicit objective since Turney (2000), priced as a fixed *deployed set* (Zhou et al., 2002; Ji and Li, 2023; Maier et al., 2024; Maier and Purucker, 2026; Gunasekaran et al., 2022; Moreira et al., 2026), as *per-example adaptive execution* (Xu et al., 2013; Trapeznikov et al., 2013; Nan and Saligrama, 2017; Chen et al., 2020), or as a *continuous penalty* in a scalarized training objective (Xu et al., 2012; Zhang and Zhou, 2011); automated machine-learning systems expose inference-time constraints during ensemble construction (Erickson et al., 2020; Feurer et al., 2022; Borchert et al., 2022). For post-hoc ensembles over cached tabular predictions (our own setting), Maier et al. (2024) already return a Pareto front of accurate and efficient ensembles with inference time in the objective vector over 83 tabular classification datasets, and Maier and Purucker (2026) extend that line, name the objective *deployment cost* and report memory usage as their most effective cost metric. Inference cost as an objective of a post-hoc weighted ensemble is therefore prior art, and we claim no part of it.

The distinction we need falls between these categories. That the accounting itself can reorder a comparison is already known: Wang et al. (2022) place two cost accountings side by side (every member executed, versus early exit) and show that which committee looks best changes with the accounting. Their contrast, however, is between ensemble and cascade *architectures*, not between two cost functions applied to one and the same weight vector. Narrowed to that resolution, to the best of our literature search no prior work contrasts the step cost a deployed support pays with the continuous relaxation the optimizers minimize *for one weight vector of one weighted soft-voting ensemble*, and reports that the two disagree about which method wins. Where a related question has been settled it went the other way: for budgeted learning with feature reuse, a support-cost programme is provably solved by its linear relaxation (Nan et al., 2016). Our setting does not inherit that guarantee, and Section 5.7 shows the two definitions disagreeing.

2.7. The gap

Ensemble weights on a simplex, mixture designs over them, and Scheffé surrogates of ensemble performance all exist, and for classifier ensembles specifically they exist together in Kwon et al. (2024); adding NBI on those surrogates exists in Rocha et al. (2025) for a regression ensemble and in Bacci et al. (2019) for forecast combination. Multiobjective optimization of continuous ensemble weights exists. Cost-aware ensemble selection with deployment cost as a Pareto objective exists (Maier et al., 2024; Maier and Purucker, 2026). Metamodel-free NBI on measured objectives exists (Gellerich and Majschak, 2023), and that anchor misplacement degrades a scalarization is established (Isermann and Steuer, 1988; Deb et al., 2010; Herrmann-Wicklmayr and Flaßkamp, 2026). What does not exist, to the best of our literature search, is an evaluation of whether this pipeline can be trusted once it is both multiobjective and applied to classifiers: no external validation of the mixture surrogate on unseen compositions, no computation of the anchors both ways on one problem so that surrogate-derived and real anchors

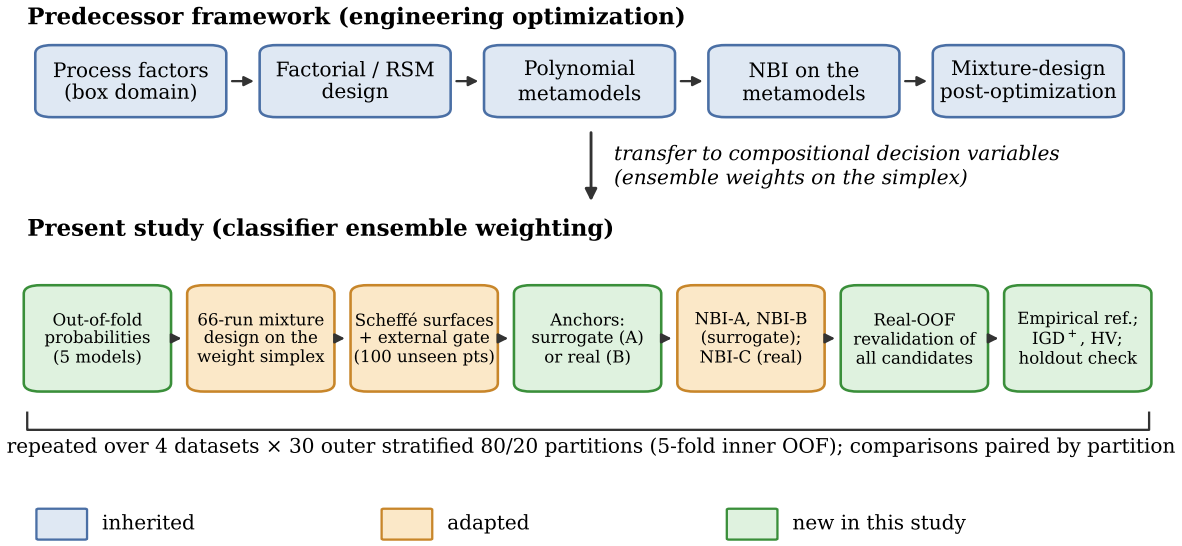


Figure 1: The predecessor DoE–RSM–NBI framework (top) and its transfer to classifier ensemble weighting (bottom). Blue: inherited. Orange: adapted because the decision variables are compositional and the responses are machine-learning metrics. Green: introduced in this study.

can be contrasted, no metamodel-free arm run alongside the surrogate arms as a control, no revalidation of candidates on the true objectives against an empirical reference whose sampled core contains no output of the compared methods, no replicated partition study of the whole pipeline, and no contrast, for one and the same weight vector, between the cost that is optimized and the cost that is paid.

The gap is therefore not a missing method. It is the absence of the controls that would tell a practitioner when the existing method's output can be believed, which is what this paper supplies, for the one-shot, no-infill form of the pipeline described above.

3. Methodology

Figure 1 shows the pipeline of this study, marking each element as inherited, adapted for compositional decision variables, or new here.

3.1. The ensemble-weighting problem is a mixture problem

Let M probabilistic base classifiers produce estimates $p_1(x), \ldots, p_M(x) \in [0,1]$ for a sample x. A weighted soft-voting ensemble predicts

$$p_{\mathbf{w}}(x) \;=\; \sum_{i=1}^{M} w_i \, p_i(x), \tag{1}$$

so that, for n samples, the vector of ensemble probabilities is $\mathbf{Pw}$, where $\mathbf{P} \in [0,1]^{n \times M}$ collects the base-model probabilities. For $p_{\mathbf{w}}(x)$ to remain a probability for every input, the weights must satisfy

$$w_i \geq 0 \quad (i = 1, \ldots, M), \qquad \sum_{i=1}^{M} w_i = 1, \tag{2}$$

that is, $\mathbf{w} \in \Delta^{M-1}$, the unit simplex in $\mathbb{R}^M$ and the canonical constraint set of a *mixture experiment*: the components are the classifiers, the vertices are single models used alone, and the centroid is uniform soft voting. The sum-to-one constraint makes an intercept model singular,

so the formalism is the mixture design with a Scheffé canonical polynomial, not a factorial design. This correspondence, rather than any performance argument, is why the framework transfers at all.

Three objectives are considered, all functions of $\mathbf{w}$ through Equation (1):

$$f_1(\mathbf{w}) = -\,\mathrm{AUC}(y, \mathbf{P}\mathbf{w}) \qquad \text{(discrimination, maximized)}, \tag{3}$$

$$f_2(\mathbf{w}) = \frac{1}{n}\sum_{k=1}^{n}\Big[-y_k \log p_{\mathbf{w}}(x_k) - (1-y_k)\log\big(1-p_{\mathbf{w}}(x_k)\big)\Big] \qquad \text{(calibration, minimized)}, \tag{4}$$

$$f_3(\mathbf{w}) = c(\mathbf{w}) \qquad \text{(inference cost, minimized)}. \tag{5}$$

The log-loss f_2 is convex and smooth in $\mathbf{w}$, whereas the area under the ROC curve is a rank statistic: f_1 is piecewise constant in $\mathbf{w}$.

3.2. Two cost models: a relaxation and a deployment cost

Let c_i be the measured inference cost of model i (Section 4.2). Deployment cost is a step function: a model receiving *any* non-negligible weight must be loaded and executed, one receiving none costs nothing, giving the *support cost*

$$c_{\mathrm{sup}}(\mathbf{w}) \;=\; \sum_{i=1}^{M} c_i\,\mathbf{1}[w_i > \varepsilon]\,, \qquad \varepsilon = 10^{-3}, \tag{6}$$

which is discontinuous; every optimizer here instead sees the continuous relaxation

$$c_{\mathrm{w}}(\mathbf{w}) \;=\; \sum_{i=1}^{M} w_i c_i, \tag{7}$$

the natural linear mixture response, which coincides with c_{sup} at the vertices but not on the interior. We therefore optimize Equation (7) and score the *same* solutions under Equation (6), reporting every comparison under both; Section 5.7 shows that the choice changes which method wins. Because each c_i is timed separately, including that model's own preprocessing, Equation (6) is the time to run the active members one after another; it excludes model loading, and parallel execution or preprocessing shared between members would give a different cost (Section 7).

3.3. Mixture design and Scheffé surrogates

Design. With $M = 5$ the design is a fixed 66-run mixture design, enumerated in supplementary Section S1, with points on the 5 pure vertices, all 10 binary edges, the ternary and quaternary faces and uniform voting: the surrogate is supported on every face a sparse ensemble could occupy.

Models. For a response y the Scheffé canonical polynomials of order 1, 2 and the special cubic are

$$\text{linear:} \quad \hat{y} = \sum_i \beta_i w_i, \tag{8}$$

$$\text{quadratic:} \quad \hat{y} = \sum_i \beta_i w_i + \sum_{i<j} \beta_{ij} w_i w_j, \tag{9}$$

$$\text{special cubic:} \quad \hat{y} = \sum_i \beta_i w_i + \sum_{i<j} \beta_{ij} w_i w_j + \sum_{i<j<k} \beta_{ijk} w_i w_j w_k, \tag{10}$$

fitted by ordinary least squares without an intercept and without pure quadratic terms, which $\sum_i w_i = 1$ makes redundant. No term elimination is performed: the canonical form is kept intact, so β_i remains the predicted response at vertex i.

External validation and the reliability gate. Each replication also evaluates 100 Dirichlet compositions outside the design, never used for fitting (supplementary Section S1), giving an external R^2_{ext} and Spearman ρ_{ext}. *Order selection* uses a parsimony rule (the lowest order whose external RMSE is within 10% of the best), so it is driven by out-of-design accuracy rather than in-sample fit. The *same* 100 compositions serve both steps, so the gate is not independent of order selection (the order it judges was chosen to do well on exactly these points) and its pass rate should be read as an optimistic estimate of out-of-design adequacy. Because R^2_{ext} is ill-posed on a nearly flat surface (its denominator is the variance of an almost constant response), Table 2 also reports the external RMSE of the selected surface in the response's own units against the observed range over the same 100 points, the more informative adequacy measure there. *Admissibility* uses a *reliability gate* whose thresholds were fixed and committed to version control before the replicated benchmark was run, and never adjusted afterwards: a surface is reliable only if

$$R^2_{\text{ext}} \geq 0.5 \quad \text{and} \quad \rho_{\text{ext}} \geq 0.9. \tag{11}$$

We describe this as pre-specification rather than preregistration: there is no external registry entry, and the evidence is the version-controlled commit history. No source is cited for the two thresholds: they are a convention chosen for this study, set before the replicated benchmark was run and not calibrated on its results, although a single-dataset Santander pilot run had been completed before they were committed. Table 2 reports the median R^2_{ext} and ρ_{ext} beside each pass count, so that their distance from the thresholds can be read directly. The gate is a hypothesis under test, not a filter applied to the results: surrogate-based optimization is run and reported *regardless* of the gate outcome, so that its predictive value can be measured (Section 5.3; supplementary Section S6).

3.4. Normal Boundary Intersection on the simplex

NBI (Das and Dennis, 1998) was introduced to obtain a distributed representation of the Pareto boundary (Section 2.4). Whether that distribution survives the map into revalidated objective space is one of the questions this paper tests, and Section 5.4 reports that in our setting it does not. For m objectives $F = (F_1, \ldots, F_m)$ the payoff matrix Φ has columns $F(\mathbf{w}^{*(k)})$ for the minimizers $\mathbf{w}^{*(k)}$ of each F_k; its diagonal is the utopia point, and objectives are normalized by the utopia and pseudo-nadir points extracted from Φ. With $\bar{\Phi}$ the normalized payoff matrix and $\hat{n} = -\bar{\Phi}\mathbf{1}$ the quasi-normal direction, each β on the unit simplex of *scalarization* weights defines the subproblem

$$\max_{\mathbf{w},\, t} \; t \quad \text{s.t.} \quad \bar{\Phi}\beta + t\,\hat{n} = \bar{F}(\mathbf{w}), \quad \mathbf{w} \in \Delta^{M-1}. \tag{12}$$

The simplex constraint is eliminated by optimizing over $M-1$ free variables with $w_M = 1 - \sum_{i<M} w_i$, with a deterministic feasibility-recovery step for iterates that leave the simplex: the free variables are clipped to $[0,1]$ and, if they still sum above one, renormalized onto the face $w_M = 0$. This is a documented repair, not a Euclidean projection onto the simplex, and a candidate recovered this way is flagged infeasible and excluded from the certified set. Without it, subproblems near a vertex terminate outside the feasible region. For vertex values of β the solution is the corresponding anchor, returned directly rather than re-derived. All variants solve 66 subproblems per replication on the same β lattice, so the arms are compared at identical points of the CHIM.

The three variants. The variants are defined by two methodological factors, the source of the objectives and the source of the anchors:

Variant	Objectives in Equation (12)	Anchors / payoff matrix	Real evaluations, standalone
NBI-A	Scheffé surfaces	optima of the Scheffé surfaces	1.7×10^2
NBI-B	Scheffé surfaces	real out-of-fold optima	$\approx 4.1 \times 10^4$
NBI-C	real out-of-fold objectives	real out-of-fold optima	$\approx 4.6 \times 10^5$

NBI-A is the lineage construction: everything, including the payoff matrix, comes from the metamodel. NBI-B changes only the anchors, which become the real optima of Section 3.5; NBI-C removes the metamodel entirely. Here and below, a *real* anchor or optimum is one computed directly on the out-of-fold objectives of the training part (Section 4.2) rather than read off a Scheffé surface. The last column charges every variant for the real evaluations it needs *standalone*, each arm carrying the stages it depends on rather than its solver stage alone (itemized in Table 5); design, gate and ROC-AUC-search entries count evaluated compositions, whereas NBI-C's solve counts each ROC-AUC and each log-loss computation separately, two per weight-vector evaluation. Real anchors are not free: NBI-B consumes about 4×10^4 real evaluations that NBI-A does not (the direct ROC-AUC search of Section 3.5), and its standalone wall clock is 1.2–12.7 times NBI-A's. Because f_1 is piecewise constant, NBI-C uses finite-difference steps of 10^{-3}, multistart on the simplex, and accepts an iterate whose equality residual is below 10^{-3} even without certified optimality, whereas NBI-A and NBI-B optimize smooth polynomials and require full certification by sequential least-squares quadratic programming (SLSQP; Kraft, 1988), as implemented in SciPy (Virtanen et al., 2020). This asymmetry follows from the objective, not from tuning, and it bounds the causal reading. A versus B is not a single-factor contrast: replacing the anchors simultaneously moves the utopia point, rescales both normalization axes through the payoff matrix, reorients the CHIM and every quasi-normal direction, and, because the three vertex values of β return their anchor directly, substitutes three returned candidates. Section 5.3 separates the last of these from the rest. B versus C tests removal of *surrogate error in the interior* once real anchors are supplied, but is not a perfectly identical numerical experiment, because changing the objective from a smooth polynomial to a piecewise-constant rank statistic necessarily changes solver semantics. Solver outcomes are therefore reported separately for each arm (supplementary Section S7).

3.5. References, comparators and revalidation

Single-objective references. Each replication computes seven single-objective references, tabulated in supplementary Section S2: best single model, uniform voting, logistic stacking, the SLSQP log-loss optimum (convex, hence global), a derivative-free ROC-AUC optimum over 4×10^4 sampled compositions with local refinement, and the two Scheffé-surface optima. The log-loss and AUC optima double as the real anchors of NBI-B and NBI-C. SLSQP on the convex log-loss remains the correct tool for the single-objective problem; NBI is not proposed as a replacement for it.

Comparators. Two cheap alternatives run on every replication: random weighted scalarization, which draws 66 scalarization weight vectors from Dirichlet($\mathbf{1}$) and minimizes the weighted sum of the *same* Scheffé surfaces with the same real anchors as NBI-B; and a 66-point random Dirichlet($\mathbf{1}$) sample of the weight simplex. The second is matched to the NBI sets only in the *number of returned candidates* and is not compute-matched; it is a floor check on the geometry of the reference front, not an efficiency baseline.

Revalidation. Every candidate from every method (the three NBI arms, both comparators, the 66 design runs and the references) is re-evaluated on the exact out-of-fold objectives, and non-dominated filtering applied to those revalidated values, before any indicator is computed. No result in this paper is read off a surrogate.

3.6. Empirical Pareto reference and quality indicators

The true Pareto set of Equation (1) over Δ^{M-1} is unknown, so each replication constructs an *empirical Pareto reference*: at least 10^5 compositions from Dirichlet($\mathbf{1}$) and Dirichlet($0.3 \cdot \mathbf{1}$) in equal parts, augmented with the simplex lattice, all vertices and edges, and a 40-cap ε-constraint sweep, evaluated on the real objectives and filtered to the non-dominated set. An independent

sample of 2×10^4 points checks convergence: if it displaces more than 5% of the front, the reference is enlarged and rechecked, up to three rounds (per-dataset convergence diagnostics in supplementary Section S8). This sampled core contains no output of the five methods compared in this study (NBI-A, NBI-B, NBI-C, random weighted scalarization and NSGA-II). It is not, however, free of optimization output: it carries the 40 ε-constraint solutions and the seven single-objective references of Section 3.5, among them the real ROC-AUC and log-loss optima that NBI-B and NBI-C receive as anchors and the two Scheffé-surface optima that NBI-A receives. On Santander the ε-constraint sweep supplies a median of 38 of the roughly 135 core-front points and the single-objective references a median of 4 (`tables/reference_diagnostics.csv`). The final reference used for scoring is the non-dominated union of that core with every candidate set, making it at least as good as any method scored against it.

We call this an empirical reference, never a true Pareto front: it is a sample-based upper bound, and because candidate sets contribute to it, a method contributing many points is partly graded against itself.

Approximation sets are scored on min–max normalized objectives. The two primary endpoints are IGD^+ (Ishibuchi et al., 2015) and the exact hypervolume with reference point 1.1 on each axis, expressed as a ratio to the reference hypervolume (Zitzler and Thiele, 1999); generational distance (Van Veldhuizen and Lamont, 2000), inverted generational distance, Schott spacing (Schott, 1995), coverage and the joint non-dominated fraction (Zitzler et al., 2003) are defined and reported for every set in supplementary Section S4.

3.7. The normalization box

Every indicator is computed on objectives mapped into $[0, 1]$ by one box, and the box is not a fixed property of the problem. Let $\mathcal{F}$ be the empirical reference front of the replication (the sampled core united with every candidate set) under one of the two cost definitions. The box is the componentwise minimum and maximum of $\mathcal{F}$:

$$\ell_k \;=\; \min_{F \in \mathcal{F}} F_k, \qquad u_k \;=\; \max_{F \in \mathcal{F}} F_k, \qquad \tilde{F}_k \;=\; \frac{F_k - \ell_k}{u_k - \ell_k}, \qquad k = 1, 2, 3. \tag{13}$$

Three consequences follow, and each constrains how the numbers in Section 5 may be read.

(i) Within a replication the box is common to every arm. All candidate sets of a replication are scored under the same ℓ and u, so the paired contrast between two arms is a difference of commensurable quantities and no arm is scored on a box of its own. The contrast is not, however, invariant to the box: the hypervolume ratio depends on the box only through the raw reference point $u + 0.1\,(u - \ell)$, and IGD^+ is computed on axes rescaled by $1/(u_k - \ell_k)$, so a different box can change a contrast's magnitude and, between sets that do not dominate one another, its sign. Nor is the box independent of the arms, whose candidates can dominate the core points that set a bound (Section 5.7).

(ii) The reference, and with it the box, changes when a method is added, because a new candidate set can add points to $\mathcal{F}$ and remove points it dominates; this changes the reference hypervolume and the IGD^+ reference set and can widen or narrow the box. Adding the NSGA-II arm moves median levels slightly: on identical candidate sets NBI-C's median hypervolume ratio is 0.989, 0.983, 0.982 and 0.976 without NSGA-II in the reference (Section 5.4) and 0.990, 0.981, 0.981 and 0.977 with it (Table 6). Single partitions move more than the medians, and a paired contrast close to zero can change direction when a method is added. Absolute indicator levels, and paired differences close to zero, are therefore only interpretable together with the set of methods that built the reference; for this reason the NSGA-II comparison is also scored against the sample core, which contains no output of the five compared methods and does not change when one is added (Section 5.5).

(iii) The box differs between replications, since each replication has its own reference front. Absolute levels are comparable within a replication but not across partitions, and a mean of ΔHV

Table 1: Datasets. Rows used, number of raw features, positive-class prevalence and training/holdout sizes of each outer partition (stratified 80/20). Kaggle test labels are never used; all selection uses 5-fold out-of-fold predictions within the training part. †200,000-row stratified subsample of the 595,212 available rows, drawn once with a recorded seed.

Dataset	Source	Rows	Features	Prevalence	Train / holdout rows
Santander	Santander Customer Transaction (Kaggle 2019)	200,000	200	0.100	160,000 / 40,000
BNP Paribas	BNP Paribas Cardif Claims (Kaggle 2016)	114,321	131	0.761	91,456 / 22,865
Porto Seguro	Porto Seguro Safe Driver (Kaggle 2017)†	200,000	57	0.036	160,000 / 40,000
UCI credit	Default of credit card clients (UCI id 350)	30,000	23	0.221	24,000 / 6,000

over 30 partitions is a mean over 30 different scales. Each paired difference is computed within a single box, which is why medians, win counts and per-partition distributions are reported alongside means, although a partition with an unusual box can still change a win count through consequence (i); Section 5.7 withdraws one earlier between-partition reading that this rescaling, not the optimizers, produced.

4. Experimental protocol

The computational protocol (partitions and seeds, base models, design, surfaces and gate thresholds, optimizers and scoring) was committed to version control before the replicated study was launched with the planned ten replications per dataset. Its only change during execution was a solver-robustness fix after a crash in replication 0; the replication-0 stages completed before the fix were kept, and the part of the change that can be checked on the stored surfaces moves no stored solution by more than 10^{-10} in any weight (supplementary Section S1). After the ten-replication analysis had been completed and reported, the study was extended to 30 replications, reusing the computational protocol without modification, and Section 5.9 reports what that extension changed. The NSGA-II baseline (Section 5.5) was added after the 30-replication analysis, with its configuration committed before its first run.

4.1. Datasets

Four public binary tabular classification problems span prevalence, dimensionality and feature type (Table 1): Santander (Kaggle, 2019) (high-dimensional, anonymized, numeric, 10% positives); BNP Paribas (Kaggle, 2016) (numeric plus high-cardinality categorical features, 76% positive); Porto Seguro (Kaggle, 2017) (strongly imbalanced at 3.6% positives, subsampled to 200,000 rows by a single stratified draw with a recorded seed to bound runtime); and UCI credit default (Yeh, 2009) (small and low-dimensional). For the three Kaggle competition datasets the competition test labels are not public and were never used: all training, weighting, thresholding and model selection use only the published training file, from which every performance figure in this paper is derived.

4.2. Base models and leakage control

The ensemble contains $M = 5$ deliberately heterogeneous learners with fixed hyperparameters and no hyperparameter search: ℓ_2-regularized logistic regression; Gaussian naive Bayes; k-nearest neighbours with $k = 100$; a random forest with 200 trees and a minimum of 20 samples per leaf; and gradient-boosted trees (XGBoost, 400 rounds, depth 6, learning rate 0.1, subsample and column subsample 0.8) (Chen and Guestrin, 2016). The first four are the `scikit-learn` implementations (Pedregosa et al., 2011). These values were committed to the project code on 2026-07-06, before the replicated study, and were not changed afterwards; Gaussian naive Bayes uses the library defaults, as do the inverse regularization strength of logistic regression ($C = 1$) and the XGBoost depth. The zoo spans the accuracy–cost–calibration space rather than forming a competitive stack: Gaussian naive Bayes is the best member out of fold on Santander but has

the worst log-loss on the other three datasets, and k-nearest neighbours, the weakest member by ROC-AUC on Santander and Porto Seguro, costs more at inference time than the other four combined (supplementary Section S2; implications in Section 7). Preprocessing (imputation, standardization and encoding, detailed per model in supplementary Section S1) is fitted inside each training fold and never on validation data.

Each replication splits the dataset by a stratified 80/20 partition into a training part and an untouched holdout. Within the training part, stratified 5-fold cross-validation produces out-of-fold probability estimates $\mathbf{P}$ for all five models; these are the only quantities the design, the surfaces, the optimizers and the selection rules ever see. Each model is then refitted once on the whole training part to produce holdout probabilities $\mathbf{Q}$, used exclusively for the confirmation analysis of Section 5.8. Every entry of $\mathbf{P}$ and $\mathbf{Q}$ is clipped to $[10^{-3}, 1 - 10^{-3}]$, a constant fixed before the study and never changed: k-nearest neighbours with $k = 100$ outputs exact zeros and ones, which can make the log-loss infinite at its vertex, and clipping the components rather than the blend keeps Equation (1) linear in $\mathbf{w}$. Because Equation (1) is linear in $\mathbf{w}$, evaluating any weight vector costs one matrix–vector product, which is what makes an empirical reference of 10^5 points and a metamodel-free NBI arm affordable.

Inference cost c_i is measured, not assumed: per model and replication, the median wall-clock time of five consecutive `predict_proba` calls of the refitted model, including preprocessing, in milliseconds per 1,000 rows. One batch of holdout rows, drawn with the replication seed, is shared by the five models (10,000 rows; the whole 6,000-row holdout on UCI credit). The models are timed one after another in a fixed, non-randomized order (logistic regression, Gaussian naive Bayes, k-nearest neighbours, random forest, XGBoost), each right after its own refit, an untimed holdout prediction and an untimed warm-up call on the batch. Three of them (k-nearest neighbours, the random forest and XGBoost) are set to use all 14 cores (`n_jobs` $= -1$); the code and the launch script set no other parallel option or thread limit for the timed calls.

4.3. Replication, seeds and the statistical unit

The entire pipeline, from the outer partition through scoring, is executed $R = 30$ times per dataset, giving 120 replications, with the outer partition, the inner folds and every stochastic component seeded by $20260904 + r$ for replication r. All seeds are persisted in the manifest, and the checkpointed runner described in the Reproducibility statement recomputes any stage exactly, except that re-running the out-of-fold stage re-measures the wall-clock costs c_i and therefore changes every result that depends on them.

The statistical unit is **one outer partition of one dataset**. Because all methods are evaluated on the same partition, every comparison is *paired*. Two consequences constrain the analysis throughout.

First, the 30 partitions of a dataset overlap heavily: two independent stratified 80/20 draws have an expected intersection of $0.8 \times 0.8 = 64\%$ of the dataset, which is 80% of each training part, a Jaccard index of two-thirds. Any two training parts therefore share most of their rows. The replication distribution therefore measures *sensitivity to data partitioning*, not sampling from a population, and the effective information is far below 30 independent runs. Second, and consequently, the dataset is the unit of generalization: results are summarized per dataset and the four dataset-level results are then compared. Nothing is pooled across the 4×30 replications, and no statement of the form "$N = 120$ independent experiments" is made anywhere in this paper.

4.4. Statistical analysis

The statistical analysis plan (supplementary Section S1) designates four primary comparisons. NBI-B versus NBI-A and NBI-C versus NBI-B address RQ4 and RQ5. NBI-C versus random scalarization and NBI-C versus the random Dirichlet sample compare NBI-C, as specified and not at equal budget, with the two cheap alternatives of Section 3.5: the first comparison is

confounded with the objective source and the second is a floor check. The plan was committed to version control after the first ten replications had been run and analysed, minutes after the extension to 30 replications was launched and before any of its partitions had been completed, so it is prospective only for partitions 10–29. The paired contrasts coded before the benchmark produced its first output were descriptive and compared both cheap alternatives with NBI-B rather than NBI-C; Section 5.4 reports both pairings. The primary endpoints are IGD^+ and the hypervolume ratio, computed on the revalidated *out-of-fold objectives of the 80% training part* (the only quantities the design, the surfaces, the optimizers and the selection rules ever see; Section 4.2) against the empirical reference of the same replication, under the weighted cost that the optimizers actually minimized; the support-cost repetition is a sensitivity analysis. The untouched 20% holdout enters no primary endpoint: it is used only for the confirmation analysis of Section 5.8. All other indicators are descriptive.

Paired differences are signed so that $\Delta > 0$ always favours the first-named (new) method: $\Delta\mathrm{HV} = \mathrm{HV}_{\mathrm{new}} - \mathrm{HV}_{\mathrm{ref}}$ and $\Delta\mathrm{IGD}^+ = \mathrm{IGD}^+_{\mathrm{ref}} - \mathrm{IGD}^+_{\mathrm{new}}$. For each dataset $\times$ comparison $\times$ endpoint we report the mean and median with percentile-bootstrap 95% intervals (10,000 resamples, fixed seed), the quartiles and range (Efron and Tibshirani, 1993), the win/tie/loss counts with a tie tolerance of 10^{-4}, the win fraction with a Wilson 95% interval (Wilson, 1927), and the matched-pairs rank-biserial correlation (Kerby, 2014) as a distribution-free effect size. These and all other replication-level intervals in this paper treat the replications of a dataset as independent observations: they describe partition sensitivity within that dataset, carry no correction for partition overlap and do not by themselves establish significance.

An ordinary paired t-test on resampled partitions is anticonservative, so the primary test is the Nadeau–Bengio corrected resampled t-test (Nadeau and Bengio, 2003): with $J = 30$ paired differences d_j,

$$t = \frac{\bar{d}}{\sqrt{\mathrm{var}(d)\left(\frac{1}{J} + \rho\right)}}, \qquad \rho = \frac{n_{\mathrm{test}}}{n_{\mathrm{train}}} = \frac{0.20}{0.80} = 0.25, \tag{14}$$

on $J - 1$ degrees of freedom. The correction inflates the variance by a factor of $(1/30 + 0.25)/(1/30) \approx 8.5$, multiplying standard errors by about 2.9 relative to the naive test. One caveat attaches to ρ and we state it plainly: the heuristic $\rho = n_{\mathrm{test}}/n_{\mathrm{train}}$ was derived for a statistic measured on the held-out portion of each resample, whereas both primary endpoints here are measured on the training part, whose rows two replications share to the extent computed above. The value $\rho = 0.25$ therefore likely understates the dependence between replications and the corrected test is, if anything, still anticonservative; every significance statement below should be read as holding *at* $\rho = 0.25$, with the effect consistency that does not depend on ρ reported alongside it. Within each dataset the eight primary tests form one family and receive a Holm correction (Holm, 1979); no correction is applied across datasets, because datasets are not pooled. A Wilcoxon signed-rank test (Wilcoxon, 1945) is reported alongside as a distribution-free check, but it is not corrected for overlap and cannot rescue a non-significant corrected result.

Where a paired-difference distribution is visibly bimodal or heavy-tailed, the mean-based test is uninformative by construction and we say so, reporting the median with its bootstrap interval, the win fraction with its Wilson interval and the rank-biserial correlation as descriptive evidence of consistency instead. Throughout, effect consistency takes precedence over p-values. Frequencies over replications are reported with Wilson and Jeffreys 95% intervals (Brown et al., 2001).

5. Results

Results are organized by research question, with per-claim provenance in the supplementary claims map. Indicators use the weighted cost of Equation (7) unless stated otherwise.

One fact frames the rest: weighting helps, but modestly. The log-loss optimum improves out-of-fold ROC-AUC over the best single model by 0.0037, 0.0056, 0.0060 and 0.0007, one to

Table 2: Surrogate reliability over 30 partitions, all quantities measured on the *same* 100 unseen compositions: the parsimony rule selects the Scheffé order on them (lowest order within 10% of the best external RMSE) and the pre-specified reliability gate ($R^2_{\text{ext}} \geq 0.5$ and Spearman $\rho_{\text{ext}} \geq 0.9$) is then scored on them, so the gate is not independent of order selection. Orders are abbreviated lin./quad./sp. cubic. Pass counts carry Wilson 95% intervals, and the median external R^2 is shown with the bootstrap interval of the mean external R^2. The last two columns give the median external RMSE of the selected surface in the response's own units and the median observed range of that response over the same 100 points: on a flat surface R^2_{ext} is ill-posed, so the RMSE read against the range is the more informative adequacy measure. Sources: `statistics/reliability_gate_r30.csv`, `tables/scheffe_orders.csv`.

Dataset	Response	Selected order (count)	Pass	Wilson 95%	Median R^2_{ext} [95% CI]	Median ρ_{ext}	RMSE_{ext}	Range
Santander	ROC-AUC	quad. 26, lin. 4	0/30	[0.00, 0.11]	-0.333 [-0.559, -0.199]	0.830	0.0087	0.040
Santander	log-loss	quad. 29, sp. cubic 1	30/30	[0.89, 1.00]	0.971 [0.967, 0.973]	0.994	0.0028	0.081
BNP Paribas	ROC-AUC	lin. 22, quad. 7, sp. cubic 1	5/30	[0.07, 0.34]	0.118 [-0.034, 0.244]	0.908	0.0096	0.053
BNP Paribas	log-loss	lin. 21, quad. 7, sp. cubic 2	3/30	[0.03, 0.26]	-0.180 [-0.246, 0.042]	0.955	0.1214	0.694
Porto Seguro	ROC-AUC	quad. 26, sp. cubic 2, lin. 2	16/30	[0.36, 0.70]	0.519 [0.482, 0.581]	0.924	0.0037	0.030
Porto Seguro	log-loss	lin. 20, quad. 8, sp. cubic 2	12/30	[0.25, 0.58]	0.506 [0.492, 0.580]	0.995	0.0582	0.448
UCI credit	ROC-AUC	quad. 20, sp. cubic 10	30/30	[0.89, 1.00]	0.992 [0.989, 0.992]	0.995	0.0015	0.077
UCI credit	log-loss	quad. 27, sp. cubic 3	30/30	[0.89, 1.00]	0.993 [0.991, 0.993]	0.993	0.0066	0.427

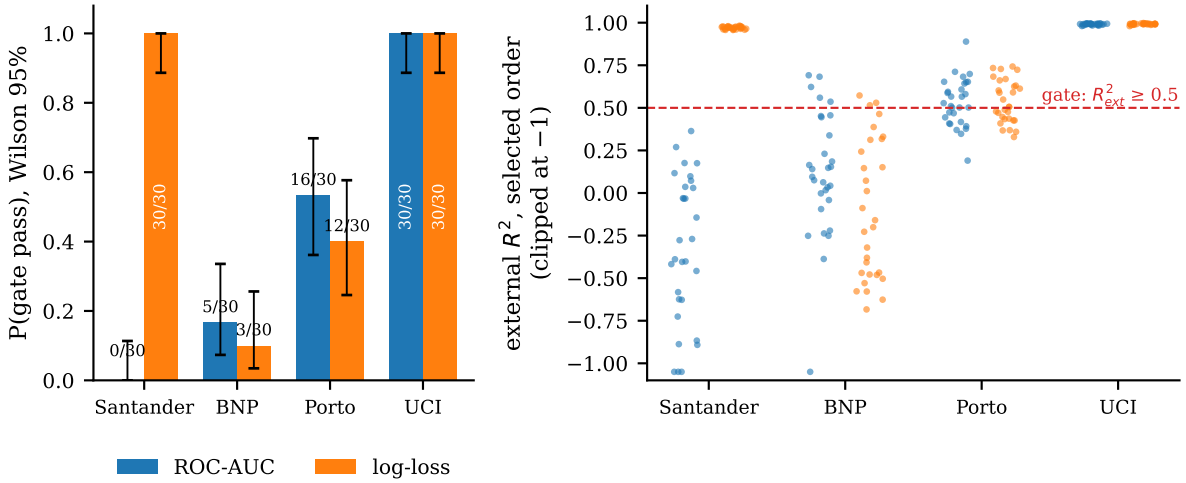


Figure 2: Left: reliability-gate pass frequency per dataset and response over 30 partitions, with Wilson 95% intervals. Right: the external R^2_{ext} of the selected order per partition; the dashed line is the gate's R^2_{ext} threshold, and because the gate also requires $\rho_{\text{ext}} \geq 0.9$, some Porto Seguro points above it fail. Within Santander the two responses fall on opposite sides of the gate.

two orders of magnitude more than the differences between the multiobjective methods studied here.

5.1. RQ1: Accuracy of the Scheffé surfaces

Surrogate adequacy depends on dataset and response alike (Table 2, Figure 2). The ROC-AUC surface passes the gate in 0/30 partitions on Santander, 5/30 on BNP Paribas, 16/30 on Porto Seguro and 30/30 on UCI credit; the log-loss surface passes in 30/30, 3/30, 12/30 and 30/30. Santander is sharpest: on the same partition its log-loss surface is essentially exact while its ROC-AUC surface has *negative* external R^2_{ext} (worse than predicting the mean) even though its external rank correlation is a respectable 0.830. Porto Seguro sits on the threshold for both responses and 30 partitions do not resolve it either way: a property of the problem, not a sample-size deficiency.

The failures are misspecification, not extrapolation: the design contains all five vertices, so every validation composition lies inside the convex hull of the design points. As an implementation check, the Brier score, which is exactly quadratic in $\mathbf{w}$, is reproduced with $R^2_{\text{ext}} = 1.000$ in

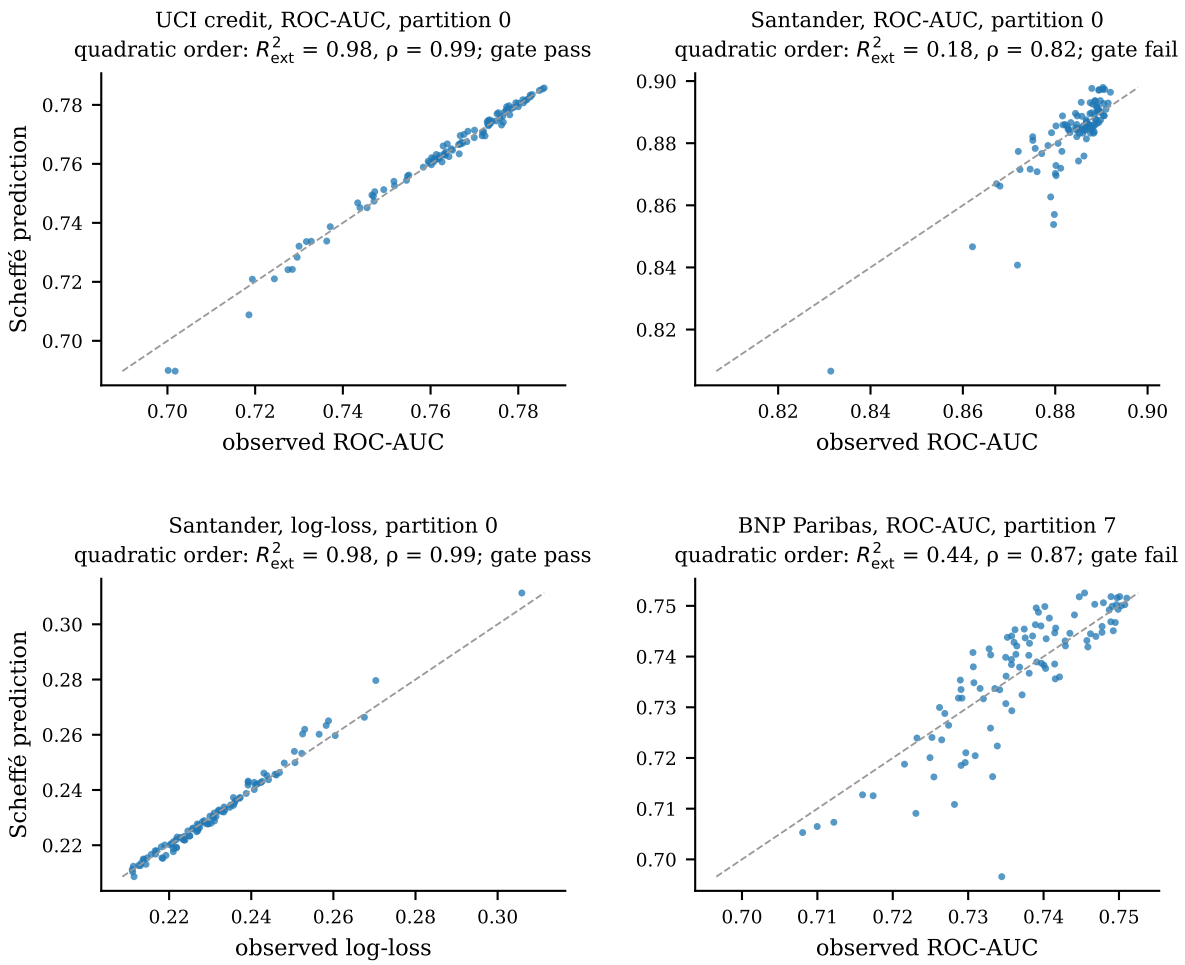


Figure 3: Predicted versus observed response on the 100 unseen Dirichlet compositions of one partition, for four dataset–response pairs spanning the regimes of Table 2. The dashed line is the identity.

all 120 replications; this confirms the fitting code and the estimability of the quadratic model on this design, not the adequacy of the other surfaces (Figure 3).

5.2. RQ2: Surface stability and what the coefficients mean

The quadratic surfaces, fitted in every partition whatever order is selected there, are stable across resampling: the four largest interaction coefficients keep their sign in all eight dataset–response cells, the largest-magnitude term is the same at $R = 10$ and $R = 30$ in all eight and is the per-partition largest in seven of them, and its between-partition coefficient of variation lies between 0.007 and 0.17 (per-cell detail in supplementary Section S3). These statistics measure sensitivity to data partitioning, not accuracy: the same 66-run design is used in every replication and any two training parts share most of their rows (Section 4.3), so the error of a misspecified surface, including any bias in its coefficients, recurs in every partition, and agreement between partitions cannot reveal it. Santander shows the difference: the leading coefficients of its log-loss and ROC-AUC surfaces both vary by about one percent between partitions (coefficients of variation 0.007 and 0.011), yet the log-loss surface passes the gate in every partition and the ROC-AUC surface in none (Table 2).

Interpreting those coefficients requires the classical criterion (Scheffé, 1958; Cornell, 2002) stated correctly. The edge profile $\hat{y}(t) = \beta_j + (\beta_i - \beta_j)t + \beta_{ij}t(1-t)$ implied by Equation (9) has an interior optimum above both pure components if and only if

$$\beta_{ij} \;>\; |\beta_i - \beta_j| \tag{15}$$

Table 3: The classical synergism criterion applied correctly, and what the real objectives say. On the binary edge i–j of a quadratic Scheffé model the fitted surface has an interior optimum exceeding both pure components if and only if $\hat{\beta}_{ij} > |\hat{\beta}_i - \hat{\beta}_j|$. Columns 3–5 evaluate that criterion for the largest-$\hat{\beta}$ pair of each dataset; column 6 gives the partitions in which the *real* out-of-fold 50/50 blend of the same pair beats its better member. The last three columns compare the criterion with the real 50/50 blend across all ten pairs: agreements, cases where the surface predicts a superior blend that the midpoint mixture does not deliver, and the reverse. Because only the midpoint of each real edge is tested, the tally overstates over-prediction and cannot detect a superior real blend elsewhere on the edge. Source: `tables/edge_condition_summary.csv`.

		Largest-$\hat{\beta}$ pair, fitted surface			Real blend	All 10 pairs		
Dataset	Pair	$\hat{\beta}_{ij}$	$\lvert\hat{\beta}_i - \hat{\beta}_j\rvert$	criterion met	beats better member	agree	surface over-predicts	under-predicts
Santander	GNB·kNN	0.243	0.130	30/30	0/30	6/10	4	0
BNP Paribas	GNB·XGB	0.148	0.110	30/30	0/30	4/10	6	0
Porto Seguro	GNB·kNN	0.071	0.025	30/30	30/30	7/10	3	0
UCI credit	GNB·XGB	0.130	0.100	30/30	0/30	3/10	7	0

(derivation in supplementary Section S3), which already guards against the naive reading: a large β_{ij} on an edge joining two very unequal vertices need not beat the better one. Our finding is not that this criterion is unknown, but that *the fitted surfaces pass it and are still wrong.*

For the largest-β pair of every dataset Equation (15) holds in all thirty partitions (Table 3), yet on the real out-of-fold objectives the predicted superior blend materializes on one dataset only. The real 50/50 blend is *worse* than its better member in 30/30 partitions on Santander, BNP Paribas and UCI credit, and beats both members only on Porto Seguro (30/30; per-pair magnitudes in supplementary Section S3); the untouched holdout repeats this. Across all ten pairs of every dataset (Table 3) all 20 disagreements between the criterion and the real 50/50 blend are the surface predicting a superior blend that the midpoint mixture does not deliver; because the criterion concerns the best point of an edge and this test only its midpoint, the tally overstates over-prediction and cannot detect a superior real blend away from the midpoint.

The mechanism is visible in Figure 4: the largest interaction attaches to the *weakest* vertex of the pair (Table 3) and tracks the pair's vertex-quality gap (supplementary Section S3). Consistently, the pairs whose real blend does beat its better member in at least 80% of partitions are different, smaller-β pairs on every dataset except Porto Seguro (supplementary Section S3).

The criterion is correct and should be applied through Equation (15) rather than by reading the sign of β_{ij}, but on these ensembles it inherits the misspecification of the surface it is computed from and must be confirmed on the real objective, at the cost of one matrix product per pair, before a pair is called complementary. Log-loss interaction signs are uninformative: log-loss is convex in $p_{\mathbf{w}}$, so by Jensen's inequality a real blend never lies above its chord, yet the fitted surfaces give the six pairs without Gaussian naive Bayes stable *positive* log-loss interactions on BNP Paribas and Porto Seguro (supplementary Section S3 and `statistics/coefficient_stability_r30.csv`). The ROC-AUC signs are interpretable, but what the blend test contradicts is not their sign: it is the reading of a large β_{ij}, through Equation (15), as a blend that beats its better member.

5.3. RQ3 and RQ4: Surrogate versus real anchors

Replacing the surrogate-derived anchors with anchors computed from the real single-objective optima, changing nothing else, improves both primary endpoints in all 30 partitions on Santander (median $\Delta\text{HV} = +0.196$, median $\Delta\text{IGD}^{+} = +0.125$; Holm-corrected $p = 0.004$ and 0.007) and in all 30 on BNP Paribas (medians +0.058 and +0.0146). On the other two the improvement is not unanimous and the endpoints differ: Porto Seguro 23/30 partitions on IGD^{+} (median +0.0296) and 24/30 on hypervolume (median +0.076), UCI credit 24/30 on IGD^{+} (median +0.0499) and 24 wins with one tie and five losses on hypervolume (median +0.092). Bootstrap intervals for every median are in Table 4 (Figure 5, Figure 6).

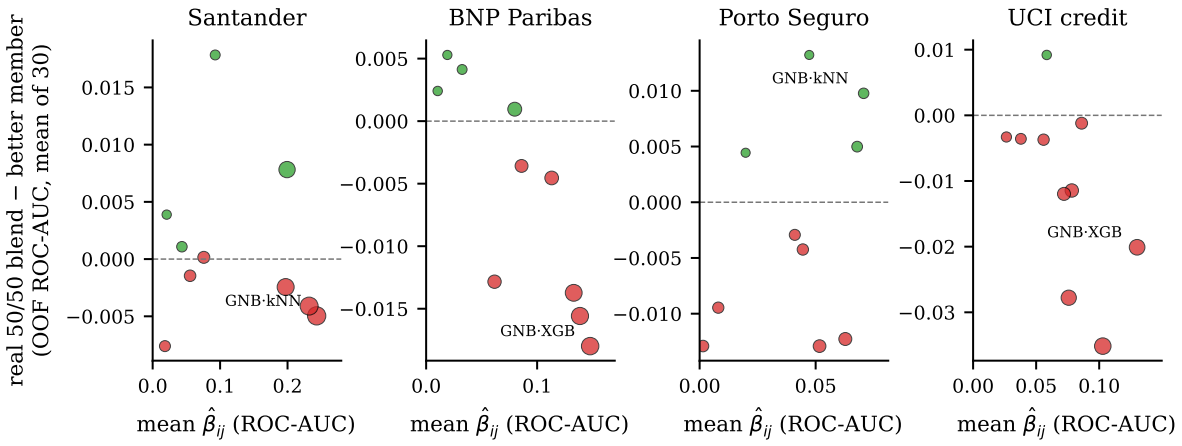


Figure 4: Fitted interaction coefficient against the real gain of the corresponding 50/50 blend over its better member, one point per classifier pair, averaged over 30 partitions; marker size is the vertex-quality gap. Green markers denote pairs whose blend wins in at least 80% of partitions; the largest-β pair is labelled.

Two readings the win counts invite are not supported by the magnitudes. First, a unanimous count is not a large effect: Santander and BNP Paribas are both 30/30, yet the median hypervolume gain is +0.196 on the first and +0.058 on the second. Second, the benefit does not track surrogate reliability: ranked by median hypervolume gain the datasets are Santander (ROC-AUC gate 0/30), UCI credit (30/30), Porto Seguro (16/30) and BNP Paribas (5/30), so the dataset whose ROC-AUC surface never passes the gate shows the largest effect and the dataset whose surfaces always pass shows the second largest. Real anchors helped on all four datasets, including the one whose surfaces are excellent; what is confined to the unreliable end is the collapse mode described below, not the typical magnitude of the benefit.

Table 4 scores every set against the augmented reference of its own replication (Section 3.6). The sample-core repetition of all four primary comparisons, computed from `nsga2/nsga2_pareto_quality.csv`, moves the magnitudes without changing the direction of any NBI-B versus NBI-A comparison and leaves the ordering above inverted in the same way; it does reverse two small IGD^+ readings, discussed in Section 5.4 (supplementary Section S5).

Three qualifications bound this. First, corrected significance is the exception here, not the rule. Of the sixteen primary tests that carry the two NBI comparisons (two comparisons × two endpoints × four datasets, weighted cost), five reach Holm-corrected $p < 0.05$: both Santander anchor tests, NBI-C versus NBI-B on Santander hypervolume, and both UCI credit NBI-C tests (Table 4). The other eleven do not: every test on BNP Paribas and Porto Seguro, both UCI credit anchor tests, and Santander IGD^+ for NBI-C versus NBI-B. On BNP Paribas the effect is unanimous in sign yet the corrected p is 0.66, because the differences are bimodal (the mean +0.279 is over four times the median +0.058), so by the rule of Section 4.4 we report the median with its bootstrap interval $[+0.051, +0.070]$, the win fraction with its Wilson interval and the rank-biserial correlation (supplementary Section S5). Those three quantities describe how *consistent* a difference is across the 30 partitions of a dataset; none of them corrects for the partition overlap of Section 4.4, which is what the corrected test exists to handle, so they do not establish significance and are not offered in its place. Where the corrected test does not reach 0.05 we report the comparison as consistent in direction and not statistically established. Second, the Porto Seguro and UCI credit win rates are driven by the twenty partitions added in the extension (hypervolume 18/20 and 19/20 against 6/10 and 5/10 with one tie in the first ten; on UCI credit the batch difference has an uncorrected Fisher $p = 0.009$, an exploratory comparison discussed in Section 5.9): the effect is consistent in direction across the 30 partitions but not statistically established, and the first ten partitions alone did not show it. Third, it

Table 4: Primary paired comparisons under the weighted cost, 30 partitions per dataset. $\Delta > 0$ favours the first-named (new) set ($\Delta\text{IGD}^+ = \text{IGD}^+_{\text{ref}} - \text{IGD}^+_{\text{new}}$, $\Delta\text{HV} = \text{HV}_{\text{new}} - \text{HV}_{\text{ref}}$). Median with its percentile-bootstrap 95% interval (not corrected for partition overlap), wins/ties/losses, and the Holm-corrected Nadeau–Bengio significance band ($\rho = 0.25$, family of eight tests per dataset; *n.s.* means the corrected test does not reach 0.05, which for a bimodal or heavy-tailed difference is expected and is why the median and the win count are reported). Each dataset's eight-test family also contains NBI-C versus random scalarization (confounded with the objective source) and NBI-C versus the random Dirichlet sample (a floor check); those comparisons, the means, rank-biserial effect sizes, exact p-values and the full support-cost repetition are in supplementary Section S5. Source: `statistics/paired_primary_effects.csv`, `paired_primary_tests.csv`.

		ΔIGD^+			Δ hypervolume ratio		
Dataset	Comparison	median [95% CI]	W/T/L	Holm p	median [95% CI]	W/T/L	Holm p
Santander	NBI-B vs NBI-A	+0.125 [+0.086, +0.145]	30/0/0	<0.01	+0.196 [+0.143, +0.216]	30/0/0	<0.01
	NBI-C vs NBI-B	+0.0017 [+0.0011, +0.0020]	28/0/2	n.s.	+0.0084 [+0.0072, +0.0093]	28/0/2	<0.01
BNP Paribas	NBI-B vs NBI-A	+0.0146 [+0.0136, +0.0172]	30/0/0	n.s.	+0.058 [+0.051, +0.070]	30/0/0	n.s.
	NBI-C vs NBI-B	-0.0028 [-0.0044, -0.0015]	6/0/24	n.s.	+0.0106 [+0.0087, +0.0128]	24/0/6	n.s.
Porto Seguro	NBI-B vs NBI-A	+0.0296 [+0.0193, +0.0730]	23/0/7	n.s.	+0.076 [+0.046, +0.157]	24/0/6	n.s.
	NBI-C vs NBI-B	+0.0244 [+0.0168, +0.0408]	28/0/2	n.s.	+0.074 [+0.039, +0.100]	28/0/2	n.s.
UCI credit	NBI-B vs NBI-A	+0.0499 [+0.0327, +0.0606]	24/0/6	n.s.	+0.092 [+0.039, +0.100]	24/1/5	n.s.
	NBI-C vs NBI-B	+0.145 [+0.126, +0.182]	29/0/1	<0.01	+0.275 [+0.219, +0.329]	30/0/0	<0.01

depends on the cost definition: under the support cost Santander is unchanged (30/30) and Porto Seguro stays positive on IGD^+ (29/30), but BNP Paribas weakens to 19–20/30, all ten of its hypervolume losses falling in the partitions whose support-cost normalization box is narrow (Section 5.7), and UCI credit *reverses*, NBI-B being worse in 26–27/30 partitions (full support-cost repetition in supplementary Section S5).

How much of this is the relocated geometry? Because a vertex value of β returns its anchor directly, NBI-B's returned set *contains* the three real single-objective optima, which are also points of the empirical reference. Part of the gain is therefore the injection of those extreme solutions rather than the relocated CHIM. We separate the two by rescoring NBI-A's own candidate set augmented with the same three anchors, changing nothing else (`scripts/pco213_anchor_injection_control.py`; per-partition results in supplementary Section S12). Injecting the anchors alone closes a median 80% of the A-to-B hypervolume gap on Santander, 90% on BNP Paribas, 49% on Porto Seguro and 1% on UCI credit, medians taken over the partitions in which that gap is positive (all 30 on the first two datasets, 24 on the other two). The residual, which is the part attributable to the relocated geometry, has median +0.038, +0.005, +0.021 and +0.091 hypervolume respectively over all 30 partitions.

This materially qualifies the headline. On Santander and BNP Paribas the mechanism is mostly the returned extreme points, not the CHIM; on UCI credit it is almost entirely the CHIM; Porto Seguro is intermediate. For the setting studied here, one-shot weighting of this fixed ensemble without infill, where a real evaluation is one matrix–vector product plus a sort and the direct search for the real optima is affordable (Table 5), the practical implication is unchanged and if anything simpler: compute the real single-objective optima and make sure they reach the returned set, whether by re-anchoring or by adding them. This is not a recommendation for surrogate-assisted multiobjective optimization in general: where each real evaluation is expensive, the cost of finding the real optima has to be weighed against the gain (Section 6.5). The causal story, however, must be stated at this resolution rather than as "relocating the CHIM repairs the front".

What goes wrong when anchors come from the surrogate. NBI-A's BNP Paribas hypervolume ratio collapses in seven of thirty partitions, exactly those in which the parsimony rule selects the

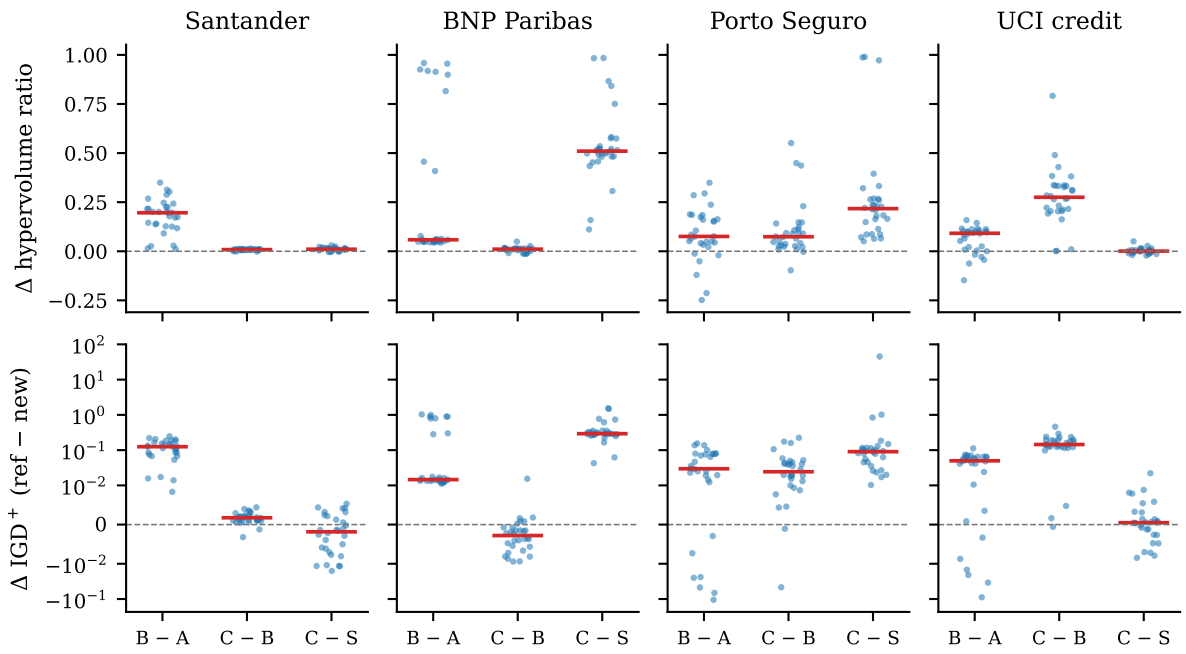


Figure 5: Paired differences on the two primary endpoints, one point per outer partition, red bar at the median; the IGD^+ axis is symmetric-logarithmic. Comparisons: B − A, NBI-B − NBI-A; C − B, NBI-C − NBI-B; C − S, NBI-C − random scalarization. The BNP Paribas NBI-B − NBI-A distribution is visibly bimodal: small positive differences plus a tail of seven partitions where NBI-A collapses.

quadratic ROC-AUC order (7/7 against 0/22 of the linear-selected ones; the remaining partition selects the special cubic order). A quadratic surface moves the surrogate AUC anchor off the gradient-boosting vertex to an interior mixture the empirical reference dominates; anchor cost tracks the collapse, but one partition collapses with a cheap anchor, making cost a marker rather than the mechanism (per-partition ratios and anchor costs in supplementary Section S6).

The consequence for the gate is uncomfortable. Because the quadratic ROC-AUC order is selected precisely when it predicts unseen points better, the collapsed partitions have *higher* ROC-AUC external R^2_{ext} (median 0.46 against 0.08) and their ROC-AUC surface passes the gate *more* often (3 of 7 against 2 of 23) than those where NBI-A works: the ROC-AUC gate does not single out this failure, while real anchors remove it, NBI-B reaching 0.935–0.996 in all thirty partitions. It also settles how the gate-conditional analysis must be read: the apparently larger anchor benefit where the ROC-AUC gate passes on BNP Paribas (+0.64 against +0.21) is NBI-A's collapse restated, since three of the five gate-passing partitions are collapse partitions, and is not evidence that real anchors add more when the ROC-AUC surface is reliable.

5.4. RQ5: Surrogate NBI against metamodel-free NBI

NBI-C gives the best or tied-best approximation among the surrogate-derived methods on every dataset (Section 5.5 shows an external evolutionary optimizer does better still). NBI-C's median hypervolume ratio is 0.989, 0.983, 0.982 and 0.976. It exceeds 0.95 in all thirty partitions of three datasets and in 26/30 on Porto Seguro (worst partition 0.877; per-partition counts in supplementary Section S4). Against NBI-B it wins on hypervolume in 28, 24, 28 and 30 of 30 partitions, at medians of +0.0084, +0.0106, +0.074 and +0.275, the corrected test reaching significance on Santander ($p = 0.003$) and UCI credit ($p = 0.010$).

Set beside the anchor effect of Section 5.3, these medians do not order the two steps the same way on the four datasets. Removing the metamodel adds far less than moving the anchors on Santander and on BNP Paribas; on Porto Seguro the two are a near-tie on the median and reverse on the mean; and on UCI credit the ordering reverses outright, removal adding three

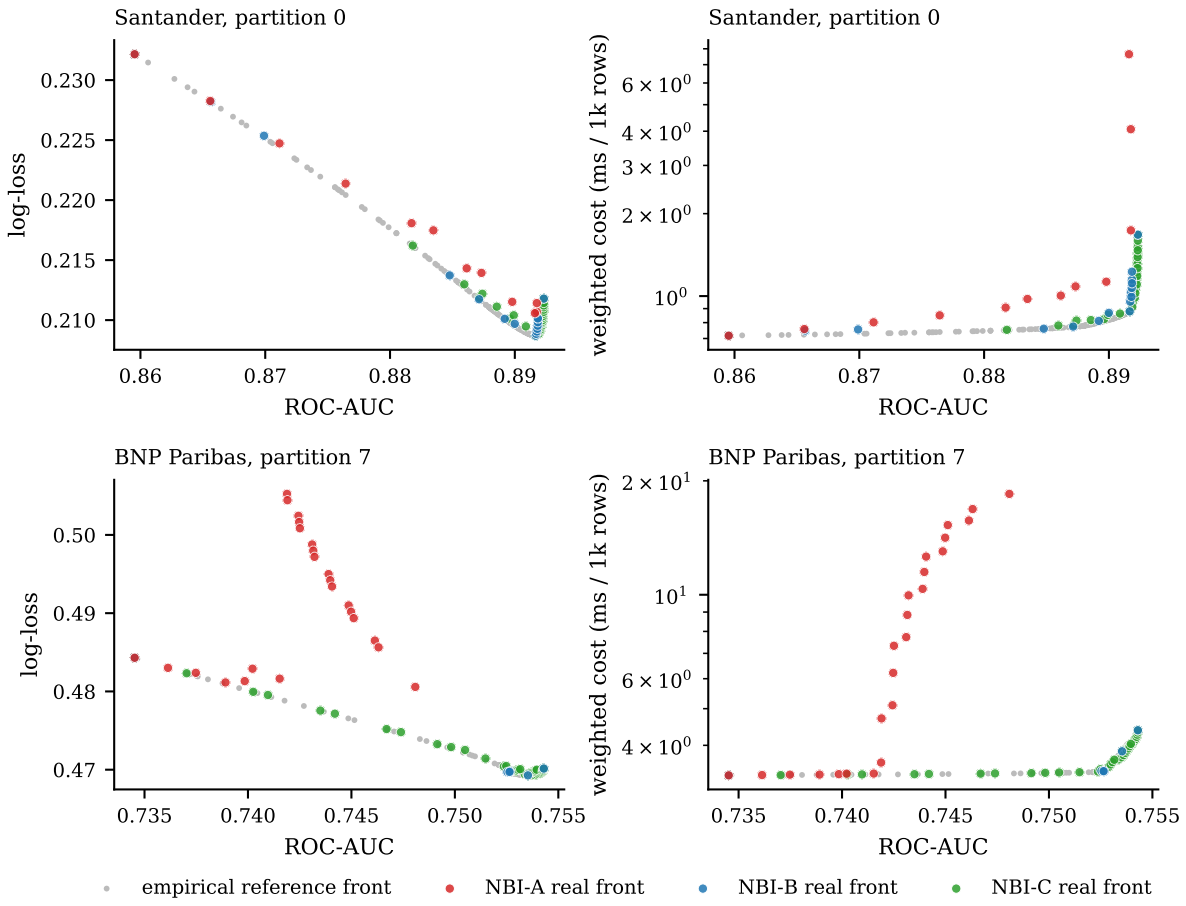


Figure 6: Revalidated fronts of the three NBI variants against the empirical Pareto reference, in the accuracy–calibration and accuracy–cost projections, for one Santander partition (top) and one BNP Paribas collapse partition (bottom). Surrogate anchors displace the NBI-A front away from the reference and up the cost axis.

times what the anchors add (supplementary Section S5). Two clean datasets, one tie and one reversal over four datasets is an observation about these problems, not a tested ordering, and it is weakened further by the confound of Section 3.4: NBI-C differs from NBI-B in the solver acceptance rule as well as in the objective source, so the *ratio* of the two effects is weaker evidence than either effect on its own.

Three caveats bound the NBI-C result. On BNP Paribas the two endpoints disagree, but the disagreement is a property of the reference definition rather than of the arms. Against the augmented reference of Table 4, NBI-C covers more volume (24/30, median +0.0106) while NBI-B is closer in IGD^+ (24/30); against the sample core, which contains no output of the compared methods, the IGD^+ comparison reverses (NBI-C is closer in 20/30) while the hypervolume direction is unchanged (24/30) (supplementary Section S5). We therefore withdraw the earlier reading of this as a genuine hypervolume/IGD^+ trade-off between the arms and record it as reference-dependent. The large UCI credit gap is not a surrogate-accuracy effect (those surfaces pass the gate in every partition) but a convergence effect of NBI-B, which certifies a median of 29 of its 66 subproblems while indicators use converged candidates only; retaining its unconverged iterates preserves the direction (30/30) but cuts the mean gap from +0.277 to +0.108 (supplementary Section S7). Neither observation supports a claim that interior surrogate error dominates once anchors are right. Third, the premium is large and uneven (Table 5): NBI-C's solve records about 4.2×10^5 real objective evaluations per replication, counting each ROC-AUC and each log-loss computation, whereas the surrogate arms' solves record none. Charged stan-

Table 5: Standalone cost of each NBI variant: mean wall-clock seconds per partition (Apple M4 Max, 14 cores) and mean real out-of-fold objective evaluations, charging every arm for everything it needs to run on its own (design, gate and ROC-AUC-search entries count evaluated compositions; NBI-C's solve counts each ROC-AUC and each log-loss computation, two per weight-vector evaluation). NBI-A is charged the 66-run design, the 100 gate compositions and the Scheffé fit ("design + fit"), since its anchors are the optima of the fitted surfaces; NBI-B is charged the same *plus* the single-objective reference stage, which is where its real anchors come from and which is dominated by the direct ROC-AUC search over 4×10^4 sampled compositions; NBI-C is charged the reference stage and its own solve, and needs neither the design nor the surfaces. Real anchors are therefore not free: NBI-B costs 1.2–12.7× the standalone wall clock of NBI-A, and the standalone premium of NBI-C over NBI-B is 6.2, 5.9, 5.9 and 2.3× (Santander, BNP Paribas, Porto Seguro, UCI credit), not the solver-stage ratio. Charging the whole reference stage to NBI-B and NBI-C is an upper bound, since that stage also computes references the arms do not consume, and the SLSQP log-loss anchor's evaluations are not separately instrumented. Not charged to any arm: the shared out-of-fold stage (335, 86, 69, 11 s) and the study's own instrumentation, namely the empirical reference, quality scoring and comparators (270, 154, 302, 74 s). "Success" is the mean fraction of the 66 subproblems that terminated successfully (A/B: SLSQP-certified; C: equality-feasible under the lenient rule of Section 3.4). Source: `tables/stage_times.csv`, `tables/nbi_runs.csv`, `references.json` and `design_points.npz` of the frozen replications.

		Stage means (s)			Standalone		Real	
Dataset	Arm	Design + fit	Single-obj. refs	NBI solve	total (s)	vs. NBI-A	evals	Success
Santander	NBI-A	4.1	—	34.0	38.1	1.0×	166	0.98
	NBI-B	4.1	444.5	34.6	483.1	12.7×	4.1×10^4	0.95
	NBI-C	—	444.5	2528.4	2972.9	78.0×	4.5×10^5	0.83
BNP Paribas	NBI-A	2.2	—	23.1	25.3	1.0×	166	0.95
	NBI-B	2.2	243.2	35.6	281.1	11.1×	4.1×10^4	1.00
	NBI-C	—	243.2	1409.8	1653.0	65.2×	4.6×10^5	0.83
Porto Seguro	NBI-A	4.0	—	84.8	88.8	1.0×	166	0.58
	NBI-B	4.0	441.2	55.8	501.0	5.6×	4.1×10^4	1.00
	NBI-C	—	441.2	2504.5	2945.7	33.2×	4.6×10^5	0.98
UCI credit	NBI-A	0.5	—	124.8	125.3	1.0×	166	0.39
	NBI-B	0.5	54.1	99.7	154.4	1.2×	4.1×10^4	0.48
	NBI-C	—	54.1	306.6	360.7	2.9×	4.7×10^5	0.98

dalone, so that each arm carries the stages it depends on, the premium of NBI-C over NBI-B is 6.2, 5.9, 5.9 and 2.3×: smallest on UCI credit, where it buys the largest gain, and largest on Santander, where it buys +0.008 hypervolume. Among the NBI variants, metamodel-free NBI is not uniformly preferable: what its premium buys depends strongly on the dataset.

Cheap comparators. Random weighted scalarization is strong on UCI credit, whose surfaces pass the gate in every partition, and on Santander, whose log-loss surface always passes it although its ROC-AUC surface never does; it collapses on BNP Paribas, whose surfaces rarely pass (Table 2), to a median of one front point and hypervolume ratio 0.474. It ties NBI-C on UCI credit (differences within ±0.002, win fractions 16/30 and 18/30, corrected $p = 1.00$). On Santander it trails NBI-C narrowly on hypervolume (0.978 against 0.989; NBI-C better in 25/30, corrected $p = 0.06$) and beats it on IGD^+ in 20/30 (corrected $p = 0.35$), but both readings depend on the reference: against the sample core, which contains no output of the compared methods, both move toward NBI-C and the IGD^+ majority reverses (computed from `nsga2/nsga2_pareto_quality.csv`; supplementary Section S5). The comparison is confounded and we say so: NBI-C optimizes exact objectives while scalarization optimizes surrogates, so its BNP Paribas and Porto Seguro advantage cannot be attributed to the NBI construction rather than to the objective source. On identical surfaces NBI-B beats scalarization 30/30 on BNP Paribas and 25/30 on Porto Seguro but loses 29/30 on UCI credit.

Table 6: NSGA-II against metamodel-free NBI (NBI-C), weighted cost, scored against the common augmented reference. Both optimize the same three real out-of-fold objectives. NSGA-II's budget was set per replication to NBI-C's count of real objective evaluations (realized ratio 0.999996 overall, 0.99992–1.00010 per run), but that count records ROC-AUC and log-loss separately, so NSGA-II performed twice as many weight-vector evaluations. Δ is the paired difference signed so that $\Delta > 0$ favours NSGA-II, with its percentile-bootstrap 95% interval, the win/tie/loss count over 30 partitions and the Holm-corrected Nadeau–Bengio p. The last column is the wall-clock ratio, which is not matched either. The direction is the same under the sample-core reference, which contains no output of the five compared methods: median ΔIGD^+ +0.0047, +0.0075, +0.0065 and +0.0034 (W/T/L 29/0/1, 30/0/0, 28/0/2, 24/0/6) and median Δ HV ratio +0.0060, +0.0098, +0.0113 and +0.0114 (30/0/0, 30/0/0, 26/0/4, 30/0/0), in row order, with the same cells reaching corrected significance. Source: `reports/pco213_postwork_benchmark/nsga2/`.

	Median HV ratio		ΔIGD^+			Δ HV ratio			Time
Dataset	NBI-C	NSGA-II	median [95% CI]	W/T/L	p	median [95% CI]	W/T/L	p	ratio
Santander	0.990	0.998	+0.0060 [+0.0039, +0.0074]	29/0/1	0.026	+0.0083 [+0.0063, +0.0110]	30/0/0	0.026	2.2×
BNP Paribas	0.981	0.994	+0.0087 [+0.0078, +0.0094]	30/0/0	<0.001	+0.0127 [+0.0113, +0.0139]	30/0/0	<0.001	2.2×
Porto Seguro	0.981	0.995	+0.0048 [+0.0035, +0.0075]	26/0/4	0.577	+0.0115 [+0.0070, +0.0169]	26/0/4	0.577	2.2×
UCI credit	0.977	0.988	+0.0032 [+0.0011, +0.0045]	23/2/5	0.131	+0.0114 [+0.0071, +0.0156]	30/0/0	0.023	2.3×

The 66-point random Dirichlet sample loses to NBI-C in every partition and to random scalarization in at least 28 of 30 on every dataset. It also loses to NBI-A and NBI-B in every partition except on UCI credit, where it beats NBI-A in 25 of 30 partitions on hypervolume and 24 on IGD^+, and NBI-B in 14 and 17 respectively. The sample's hypervolume ratio is exactly zero in up to 25/30 partitions (supplementary Section S4), because uniform compositions cost 8 to 20 times more than the reference front. The unoptimized 66-run mixture design also beats the sample 30/30 everywhere. The sample is, however, matched to the NBI sets only in the number of candidates returned, so this is a floor check and no efficiency claim is made from it.

One negative result bears on the classical motivation for NBI recalled in Section 3.4: the even spread of subproblems over the CHIM does not survive revalidation on the real objectives, under any spacing definition we computed. NBI-C's size-matched spacing percentile is 0.99, 0.54, 0.96 and 0.41, against 1.00, 1.00, 1.00 and 0.93 for the random Dirichlet sample; Section 5.5 shows this is a property of the construction, not of the objective geometry.

5.5. RQ5b: An external evolutionary baseline

Everything above compares methods within one methodological family; the external question is whether a canonical evolutionary optimizer, given the same objectives, behaves differently. We ran NSGA-II (Deb et al., 2002) in pymoo (Blank and Deb, 2020) on exactly the three real out-of-fold objectives NBI-C optimizes, with a population of 66 matching the 66 NBI directions, library-default variation operators, a deterministic simplex repair, and a termination budget set *per replication* to NBI-C's recorded count of real objective evaluations there; the configuration was fixed and committed before any run, so no operator setting was chosen by inspecting performance (supplementary Section S13). NSGA-II's evaluation counter, however, records weight-vector evaluations, each of which computes both ROC-AUC and log-loss, whereas NBI-C's records each ROC-AUC and each log-loss computation separately, two per weight-vector evaluation. The realized ratio of the two counts across the 120 replications, 0.999996 overall and within $[0.99992, 1.00010]$ per run, therefore means that NSGA-II performed twice as many weight-vector evaluations as NBI-C's solve (about 1.7 times as many if NBI-C is also charged its ROC-AUC anchor search). The comparison is matched neither in weight-vector evaluations nor in time, and the wall-clock ratio is reported separately. Because the empirical reference is augmented with candidate sets, adding a method changes it, box included (Section 3.7); we therefore score twice, against a *sample-core* reference containing no output of the five compared methods, and against a *common augmented union* that includes NSGA-II and against which every method's indicators are recomputed.

The answer is unambiguous and we state it plainly: NSGA-II attains a better approximation than NBI-C on both primary endpoints, on all four datasets, under both reference definitions (Table 6, whose caption gives the sample-core repetition). All 16 dataset × endpoint × reference cells favour NSGA-II, with rank-biserial correlations between +0.87 and +1.00. The corrected test reaches significance on Santander, BNP Paribas and UCI credit hypervolume; on Porto Seguro the paired differences are heavy-tailed (mean +0.021 against median +0.011), so by the rule fixed in advance the median, the win fraction and the effect size are the evidence there, and all three agree.

Four qualifications matter. The margins are small in absolute terms: median hypervolume-ratio gains of +0.006 to +0.013, on a scale where NBI-C already reaches 0.977–0.990. The comparison is also not cardinality-matched: NSGA-II returns exactly 66 mutually non-dominated points in all 120 runs, against NBI-C revalidated front sizes whose medians are 35, 34, 56 and 65. Hypervolume is monotone under set inclusion and IGD^+ improves with additional points near the reference, so the larger returned set mechanically helps NSGA-II on both endpoints, and within datasets the per-partition cardinality gap correlates with the hypervolume-ratio gap at Pearson +0.50, −0.37, +0.57 and +0.68 against the common augmented union. Set size cannot explain the whole margin on UCI credit, where the hypervolume gap is +0.0114 although the two sets differ by a median of one point, but elsewhere it may explain part of it, so the reported margins should be read as an upper bound on NSGA-II's advantage. NSGA-II used about 2.2× the wall clock of NBI-C's solve: twice the weight-vector evaluations at a similar time per evaluation. All NSGA-II runs except one pilot run were executed eight at a time as concurrent processes (Reproducibility statement). And NSGA-II supplies 8–29% of the augmented reference front (a mean over four probed replications per dataset; supplementary Section S8), so like NBI-B and NBI-C it is partly graded against its own output there, which is why the sample-core reference, to which it contributes nothing, is reported alongside; it gives the same answer.

This also settles what the surrogate-only comparison could not: whether NBI's lost spacing advantage reflects the objective geometry or the CHIM construction. NSGA-II's crowding-distance selection produces markedly more evenly spaced revalidated fronts on every dataset (median spacing coefficient of variation 0.74, 0.62, 0.60 and 0.65 against NBI-C's 2.28, 0.88, 1.56 and 1.92), so even spacing here is attainable and CHIM-based construction does not deliver it. Scored post hoc under the support cost, which NSGA-II never optimized, the advantage largely transfers (supplementary Section S13), most strongly on BNP Paribas; the one reversal is Santander IGD^+, where NBI-C is marginally closer in 17 of 30 partitions.

This bounds one claim: NBI-C is best or tied-best *among the surrogate-derived methods*, not among multiobjective optimizers generally. It leaves untouched every conclusion about the surrogate pipeline itself (reliability, anchor misplacement, the ROC-AUC gate's blind spot, the coefficient result, the cost-definition result), because NSGA-II is not part of that pipeline and was never proposed as a component of it. A single untuned NSGA-II configuration on the same objectives, given twice the weight-vector evaluations of NBI-C's solve, gave a modestly but consistently better approximation on these four datasets, at about 2.2× the wall clock. Whether NSGA-II also leads at an equal number of weight-vector evaluations was not tested.

5.6. RQ6: The actual multiobjective conflict

In objective space the conflict between discrimination and calibration is almost degenerate: the direct ROC-AUC optimum exceeds the SLSQP log-loss optimum by 0.00068, 0.00080, 0.00006 and 0.00034 ROC-AUC, 5 to 100 times less than the gain from ensembling at all. Two qualifications prevent the stronger statement that the problems are the same: the optima differ in weight space, the Porto Seguro pair practically coinciding (supplementary Section S9), and Santander pays 0.0032 of log-loss for its 0.00068 of ROC-AUC, so the practitioner gets a different *model* even where the objective values are nearly indistinguishable. An apparent +254 ms per 1,000 rows deployment-cost gap on Santander did not survive checking and we do not claim it: it is an artifact of the activity threshold in Equation (6) (supplementary Section S9).

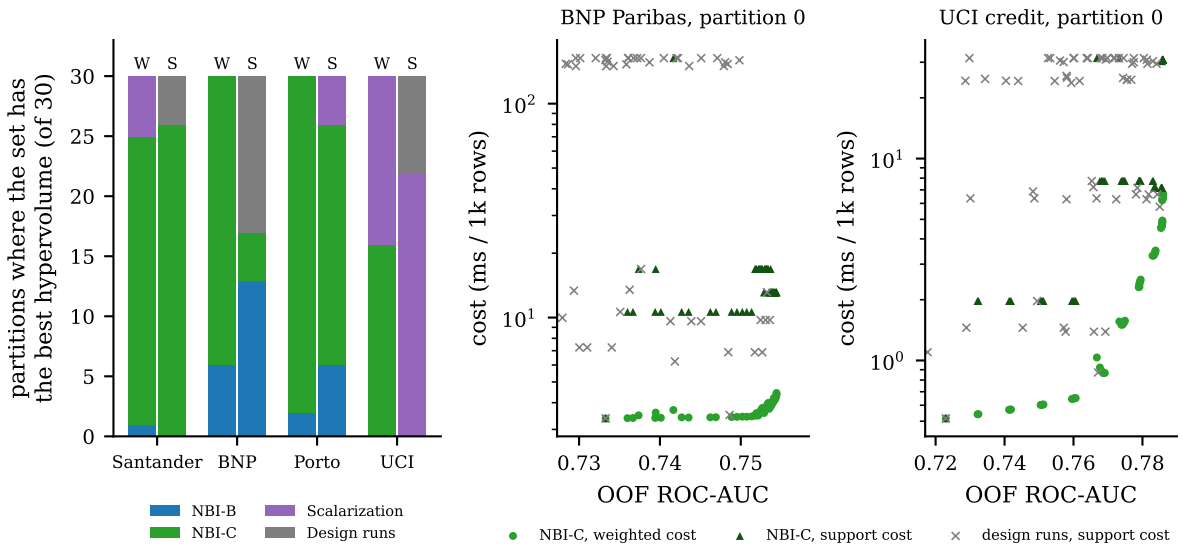


Figure 7: Left: which method attains the best hypervolume, per dataset, under the weighted cost (W) and the support cost (S), over 30 partitions. Right: one NBI-C candidate set scored under both definitions against the raw design runs, for one BNP Paribas and one UCI credit partition. Under the step cost the design runs occupy cheap supports the NBI sets, optimized on the linear relaxation, do not reach.

Table 7: Holdout ranking agreement (partitions in which the OOF-best knee pick among NBI-A/B/C and scalarization is also holdout-best, weighted cost) and cost-definition sensitivity (partitions in which the hypervolume-best set differs between the weighted and the support cost; median Spearman correlation of the set rankings). Wilson 95% intervals. Sources: `statistics/proportion_intervals.csv`, `statistics/cost_definition_sensitivity_r30.csv`.

Dataset	Holdout ranking agreement	Best set differs (W vs S)	Rank ρ (W vs S)
Santander	30/30 [0.89, 1.00]	10/30 [0.19, 0.51]	0.83
BNP Paribas	15/30 [0.33, 0.67]	24/30 [0.63, 0.90]	0.89
Porto Seguro	23/30 [0.59, 0.88]	8/30 [0.14, 0.44]	0.94
UCI credit	30/30 [0.89, 1.00]	20/30 [0.49, 0.81]	0.66

5.7. RQ7: Effect of the cost definition

The two cost definitions disagree often enough to change conclusions (Figure 7, Table 7): the hypervolume-best method changes between them in 8 to 24 of 30 partitions depending on the dataset (supplementary Section S9). Of the six candidate sets, NBI-C remains best under both on Santander and Porto Seguro (26/30 and 20/30); on UCI credit random scalarization takes the support-cost hypervolume in 22/30, and several paired readings reverse outright (Section 5.3, Section 5.4).

The mechanism, where it can be isolated, is that a method optimizing $\sum_i w_i c_i$ has no incentive to zero out a component. The raw 66-run design occupies exactly the cheap supports the NBI sets miss and can win outright under a step cost when the normalization box's cost range excludes the k-nearest-neighbour region, as it does in the 13/30 BNP Paribas partitions whose support-cost box is narrow (supplementary Section S9).

One earlier reading did not survive scrutiny and we correct it here: the apparent bimodality of NBI-C's BNP Paribas support hypervolume is a per-replication normalization artifact, not a regime. The normalization box rescales the cost axis tenfold between two groups of partitions, NBI-C's fronts are the same in both, and under a common bound the split disappears (supplementary Section S9). We therefore claim only the qualitative statement: NBI sets optimized

Table 8: Nested comparison of $R = 10$ with $R = 30$: mean hypervolume ratio and IGD$^+$ (weighted cost) of each NBI variant estimated from partitions 0–9 (the frozen $R = 10$ study, tag `pco213-postwork-r10`) and from all 30 partitions, with the $R = 30$ bootstrap interval and whether the $R = 10$ estimate lies inside it. Source: `statistics/r10_vs_r30_stability.csv`.

		Hypervolume ratio			IGD$^+$		
Dataset	Set	$R = 10$	$R = 30$ [95% CI]	inside	$R = 10$	$R = 30$ [95% CI]	inside
Santander	NBI-A	0.846	0.800 [0.767, 0.833]	no	0.0954	0.1321 [0.1078, 0.1573]	no
	NBI-B	0.982	0.980 [0.977, 0.982]	yes	0.0085	0.0121 [0.0101, 0.0142]	no
	NBI-C	0.991	0.987 [0.985, 0.990]	no	0.0069	0.0105 [0.0084, 0.0127]	no
BNP Paribas	NBI-A	0.717	0.695 [0.560, 0.819]	yes	0.2212	0.2502 [0.1234, 0.3877]	yes
	NBI-B	0.966	0.974 [0.969, 0.978]	no	0.0138	0.0114 [0.0096, 0.0132]	no
	NBI-C	0.982	0.982 [0.981, 0.984]	yes	0.0143	0.0142 [0.0134, 0.0151]	yes
Porto Seguro	NBI-A	0.803	0.781 [0.750, 0.812]	yes	0.0751	0.0911 [0.0765, 0.1055]	no
	NBI-B	0.820	0.866 [0.810, 0.913]	yes	0.0695	0.0531 [0.0351, 0.0746]	yes
	NBI-C	0.973	0.974 [0.964, 0.983]	yes	0.0113	0.0111 [0.0075, 0.0155]	yes
UCI credit	NBI-A	0.661	0.639 [0.589, 0.693]	yes	0.1854	0.2043 [0.1724, 0.2340]	yes
	NBI-B	0.670	0.699 [0.643, 0.751]	yes	0.1809	0.1681 [0.1383, 0.2008]	yes
	NBI-C	0.975	0.976 [0.974, 0.979]	yes	0.0107	0.0099 [0.0089, 0.0110]	yes

on the linear cost omit the cheapest supports the raw design contains, and the penalty's size depends on the chosen cost range.

5.8. RQ8: Transfer to the untouched holdout

In level, yes. The mean holdout-minus-out-of-fold ROC-AUC of every knee-selected solution lies within ± 0.005 as a point estimate, and the shift equals the weight-averaged shift of the base models on the same partition (correlation 0.97–1.00): a partition and refitting effect shared with the constituent models, not selection optimism.

In ranking, it depends on the dataset. The out-of-fold-best of the four knee picks is also holdout-best in 30/30 partitions on Santander, 15/30 on BNP Paribas, 23/30 on Porto Seguro and 30/30 on UCI credit, and the twenty added partitions reproduce this pattern (supplementary Section S10). The BNP Paribas figure measures how close the four candidates are, not a failure of selection: their median out-of-fold gap is 0.0005 against a holdout noise standard deviation of about 0.0012, and when the two disagree the holdout regret is at most 0.0022 ROC-AUC. These rates are specific to the selection rule and cost definition (under the support cost they become 28, 28, 17 and 16 of 30) and are not dataset properties.

5.9. RQ9: Did the first ten replications suffice?

Tripling the number of replications reversed no primary effect that is statistically established at $R = 30$, but it did not leave every direction intact. Analysed separately, the first ten and the twenty new partitions give the same sign of the median paired effect in 28 of the 32 primary cells, including all 13 that are statistically established at $R = 30$, and the same four cells are the only ones whose sign differs between $R = 10$ and $R = 30$: the UCI credit anchor effect on IGD$^+$ (real anchors winning 5/10 in the first ten partitions against 19/20 in the new ones), NBI-C versus random scalarization on Santander IGD$^+$ (NBI-C winning 6/10 against 4/20), and NBI-C versus random scalarization on UCI credit on both endpoints (medians within ± 0.0022). The level comparison of Table 8 and Figure 8 is nested (partitions 0–9 are one third of the $R = 30$ sample), so its agreement is partly built in: the $R = 10$ estimate lies inside the $R = 30$ bootstrap interval in 9 of 12 hypervolume cells, 7 of 12 IGD$^+$ cells and all 8 gate frequencies, four of which are 0/30 or 30/30 and so cannot fall outside. What the extension delivered is a change of status for three findings: the previously undecidable Porto Seguro and UCI credit anchor effect; the BNP Paribas anchor collapse, now a quantified 7/30 with an identified trigger; and two cost readings shown to be artifacts, the BNP Paribas support-cost bimodality, which

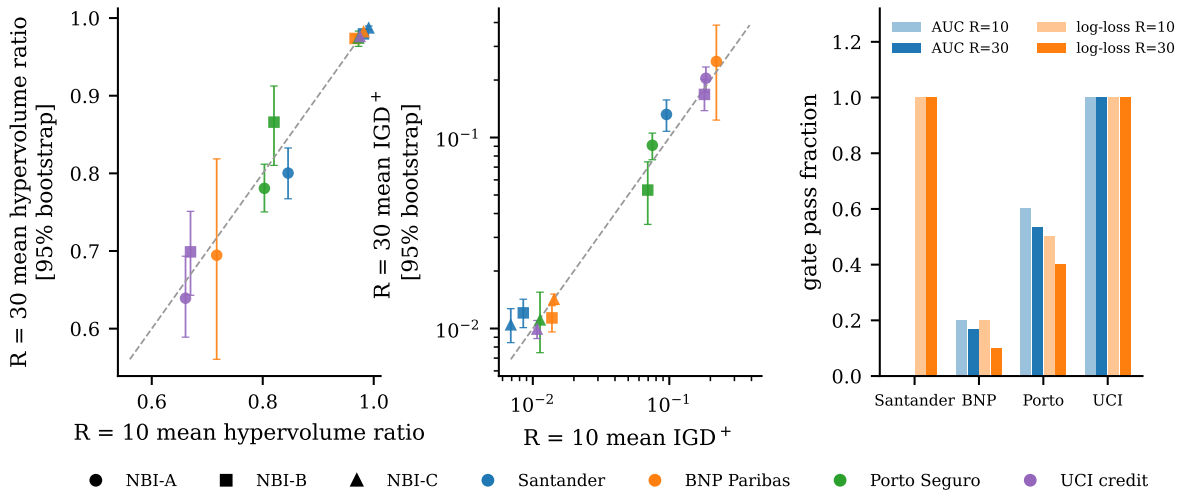


Figure 8: Mean indicator at $R = 10$ against $R = 30$ with bootstrap intervals (left, centre) and gate frequencies at both sample sizes (right). Points on the diagonal are unchanged estimates.

the ten-partition analysis had noted but not resolved, and the Santander deployment-cost gap of Section 5.6, which the ten-partition report does not record (supplementary Section S11). A fourth item on that list has since been withdrawn: the NBI-C versus NBI-B hypervolume/IGD$^+$ split on BNP Paribas is a property of the reference definition and reverses under the sample core (Section 5.4), so the extension sharpened that estimate without settling a finding.

The level check against the twenty new partitions alone is passed by only 6 of 12 hypervolume and 5 of 12 IGD$^+$ cells; the shift is clearest on Santander, whose IGD$^+$ is higher in the new partitions for all three NBI variants, and the hypervolume anchor win rates are 6/10 against 18/20 on Porto Seguro (Fisher $p = 0.14$) and 5/10 against 19/20 on UCI credit ($p = 0.009$), both uncorrected. The two batches were run on different days (partitions 0–9 on 4–6 September 2026, 10–29 on 6–8 September 2026; run conditions in Section 7), and the difference between them is unexplained. Except for the nested comparison of Table 8 and Figure 8, which the statistical analysis plan specified, the comparisons in this section are exploratory. The first ten partitions thus gave the direction of every established effect but not every conclusion (supplementary Section S11).

6. Discussion

One claim organizes the results: on the classifier-weight simplex the mixture-design response surface is a reproducible *description* of ensemble performance, broadly right in how it ranks compositions but accurate in value only on some dataset–metric pairs, and an unreliable *optimization substrate*; on Santander and BNP Paribas the dominant error is not inside the surface but in the points that build the geometry of the search (Section 5.4).

6.1. Why anchor errors are global

NBI derives its whole search geometry from the payoff matrix, so every subproblem in Equation (12) is posed relative to the individual optima. An anchor error is therefore global rather than local: it relocates the CHIM, redirects every quasi-normal and rescales the objective space of all 66 subproblems, while an interior error of the same size perturbs single solutions and leaves the frame intact.

The asymmetry predicts part of what we observe, but not all of the measured gap is geometry. Replacing surrogate anchors with real ones changes four things at once (Section 3.4), and one of them is not geometric: NBI-B's candidate set *contains* the real single-objective optima, which

are themselves points of the empirical reference. The injection control of Section 5.3 separates that channel from the rest. On Santander and BNP Paribas, where real anchors win in every partition and Santander carries the only Holm-significant anchor tests, most of the hypervolume gain comes from those returned optima; the relocated-CHIM effect, though positive in all 30 partitions of both, is small. On UCI credit the relocated CHIM accounts for almost the whole gain, and Porto Seguro is intermediate. The mechanism therefore survives at a smaller size than the raw A-to-B contrast suggests, and the implication drawn in Section 5.3 stands within its scope: in one-shot ensemble weighting with affordable real evaluations, make sure the real single-objective optima reach the returned set. Whether the same holds when real evaluations are expensive is a question of cost, taken up in Section 6.5.

How much the surrogate still costs once the anchors are right is dataset-dependent, and no single ordering covers the four: removing the metamodel entirely adds far less than real anchors did on two datasets, about the same on a third, where the median and the mean disagree, and three times as much on the fourth (Section 5.4). Over four datasets, and with the solver confound of Section 3.4, that is an observation rather than a tested claim.

BNP Paribas shows how the anchors go wrong in the first place: where the parsimony rule selects a quadratic ROC-AUC surface, the surrogate maximum sits at an interior mixture instead of the gradient-boosting vertex, so the CHIM is built from a false utopia point and the front collapses (Section 5.3). This is a geometric failure, not a numerical one, since the solver certifies these subproblems, and one that the ROC-AUC reliability gate does not single out.

6.2. Why the reliability gate does not catch it

The gate asks whether a surface predicts unseen compositions well on average; anchor placement depends on where the surface puts its *argmax*, a far more local property. A surface can be globally accurate and still misplace an extremum, or globally poor with its argmax on the right vertex.

On BNP Paribas they diverge actively: the parsimony rule selects the quadratic ROC-AUC order precisely when it predicts held-out points better, so in the collapse partitions the ROC-AUC surface has higher external R^2 and passes the gate more often than in the partitions where the surrogate anchor is right. A linear Scheffé surface has its maximum at a vertex by construction, so the *less* flexible model is accidentally protected while the more flexible one may place an interior maximum the data do not support.

Hence a division of labour: the gate is worth its 100 extra evaluations for separating unusable surfaces from excellent ones, but only the anchor check that distinguishes NBI-B from NBI-A removes the failure instead of diagnosing it.

6.3. Smooth surrogates against a rank statistic

Two of the three objectives behave as the framework's engineering origins assume: log-loss is convex and smooth in $\mathbf{w}$, and the Brier score is exactly quadratic in $\mathbf{w}$, so its surface reproduces it at $R^2_{\text{ext}} = 1.000$ in all 120 replications, an algebraic identity that checks the fitting code, not evidence about adequacy. The ROC-AUC is not: a rank statistic, constant wherever the induced ordering does not change and kinked between orderings, which a low-order polynomial smooths across instead of representing. Hence the Santander ROC-AUC surface reaches a Spearman correlation of 0.830 with a negative external R^2: ordering broadly right, values wrong.

For the leading pair on three of the four datasets, the resulting bias is over-prediction (Section 5.2): a quadratic fitted over an edge with one weak endpoint absorbs that curvature into β_{ij} and overshoots its maximum, the same over-prediction of interior optima that displaces surrogate anchors off vertices. Where the objective *is* smooth in the weights the machinery can work as advertised: Santander's log-loss surface passes the gate in every partition.

Non-smoothness is a contributing factor, not the explanation. Were it the explanation, the smooth convex objective would be the reliable one everywhere, and it is not: log-loss surfaces

pass the gate in 30/30 partitions on Santander and UCI credit but in only 3/30 on BNP Paribas and 12/30 on Porto Seguro. On BNP Paribas the smooth objective does no better than the rank statistic (3/30 against 5/30 at the pre-specified gate thresholds and the component clip of Section 4.2; median R^2_{ext} of -0.180 against $+0.118$); on Porto Seguro the two are effectively tied (12 against 16). Only Santander shows the sharp separation the smoothness account predicts (R^2_{ext} 0.971 for log-loss against -0.333 for ROC-AUC), and one dataset is an illustration rather than a mechanism. What these data establish is that adequacy varies by dataset *and* by metric and has to be measured on unseen compositions; they do not identify what, besides the shape of the response in $\mathbf{w}$, makes a given dataset–metric pair fittable by a low-order polynomial.

6.4. Non-smoothness also constrains the optimizer

The same property constrains the metamodel-free arm: SLSQP cannot certify optimality on a piecewise-constant objective, whose gradient is zero almost everywhere, so NBI-C accepts a feasible iterate under the weaker rule of Section 3.4, not the certification NBI-A and NBI-B obtain on smooth polynomials. Success rates are therefore not on a common scale across arms, and we do not read the difference as evidence about surrogate quality.

UCI credit shows why that care is needed: its $+0.277$ hypervolume gain for NBI-C over NBI-B, at Holm-corrected $p = 0.010$, looks like the strongest evidence here for the metamodel-free arm, yet UCI credit's surfaces pass the gate in every partition, so surrogate error cannot explain it: the gap tracks how many of NBI-B's 66 subproblems certify, at Spearman -0.92, and disappears in the partitions where NBI-B converges fully (supplementary Section S7).

6.5. Compute against fidelity

Among the surrogate-derived methods, metamodel-free NBI is best or tied-best on the primary endpoints, although an external evolutionary optimizer on the same real objectives, given about twice NBI-C's weight-vector evaluations, does better still (Section 5.5). Evaluating the real objectives throughout its solve, NBI-C costs 2.3 to 6.2 times the standalone wall clock of NBI-B (Table 5), and what that premium buys is strongly conditional: on UCI credit the smallest premium buys the largest gain, while on Santander the largest buys $+0.008$ hypervolume.

NBI-C's affordability is specific to this problem: one evaluation is a matrix–vector product plus a sort, which takes milliseconds here. Where each evaluation needs a model refit the ordering reverses and the surrogate is the only viable route, which is the regime the framework was designed for. The recommendation is therefore conditional on evaluation cost: cheap evaluations should be used directly, at least for the anchors; expensive ones argue for an externally validated surrogate, anchored on whatever real evaluations are affordable.

6.6. Response surfaces as descriptions rather than optimizers

The surfaces themselves are reproducible across resampled partitions (Section 5.2), although reproducibility does not make them accurate (Section 5.1), and as a description of which vertices are weak the model is interpretable in a way a black-box optimizer is not. What does not survive is the step from that description to inference about blends: the classical synergism criterion of Equation (15) is computed *from the surface*, so on these ensembles it inherits the surface's misspecification, and a candidate pair must be confirmed on the real objective.

6.7. Deployment cost is not the cost that was optimized

Every method here minimized the continuous relaxation $\sum_i w_i c_i$, since a step cost is not differentiable and cannot enter a gradient-based formulation; deployment pays the step cost. The two are not monotonically related inside the simplex, and the consequence is not cosmetic: which of the six candidate sets attains the best hypervolume changes between the definitions in 8 to 24 of the 30 partitions (Section 5.7; supplementary Section S9), and several paired conclusions reverse, including the anchor effect on UCI credit.

The mechanism is that a linear cost gives no incentive to zero a component: a weight of 0.01 on a model costing 200 ms adds 2 ms under the relaxation and 200 ms in deployment. Methods optimizing the relaxation drift into the interior and miss the cheap low-support solutions (pure vertices and two-model faces) that the unoptimized 66-run design contains by construction.

Whether a relaxation and the support it induces agree has been examined in one related setting, and the answer went the other way: for budgeted learning with feature re-use, a support-cost programme is solved exactly by its linear relaxation (Nan et al., 2016), a guarantee our setting does not inherit. Report which cost model is used; zero weights below the activity threshold and re-evaluate, since an inert weight just above ε produced a spurious difference of about 250 ms per 1,000 rows in our own analysis; and optimize over supports explicitly where deployment cost is the real objective.

6.8. What the framework's transfer teaches

Smoothness of the response in the mixture components is not enough for the pipeline to transfer: on these four datasets surface adequacy had to be measured on unseen compositions, not identified in advance (Section 6.3). Where the ROC-AUC surface is adequate too (UCI credit, which passes in 30/30), the cheap route is competitive: random scalarization on those externally validated surfaces is statistically indistinguishable from the metamodel-free arm (mean $\Delta\mathrm{HV} = +0.0015$ with bootstrap interval $[-0.003, 0.007]$, better in 16/30; indicator levels in supplementary Section S4). Surrogate-assisted NBI with real anchors is not, but that residual gap is a subproblem-convergence effect of NBI-B rather than surrogate error (Section 6.4). The stronger reason to stop early there is that there is very little to win: the gain of the log-loss optimum over the best single model is an order of magnitude smaller than on the other three datasets, out of fold and on the untouched holdout alike, and not separable from zero by a corrected test (supplementary Section S10). What UCI credit therefore shows is not that the inherited machinery is efficient but that on that dataset *no* weighting method separates from the best single model on held-out data, so the cheapest adequate one suffices. The pipeline does not transfer unexamined to a rank-statistic response, to anchors that are vertices of the feasible region rather than interior points, or to an objective that is a step function of the support.

One connection this study does not establish is between front quality and the decision taken from the front. *Within* a candidate set, out-of-fold selection transfers to the holdout in level, and the out-of-fold-best of the four knee picks is also holdout-best in 30/30, 15/30, 23/30 and 30/30 partitions (Section 5.8); across methods it is much weaker. Paired by partition under the weighted cost, the two NBI contrasts that are Holm-significant on both primary endpoints have *worse* single-objective holdout accuracy at the knee-selected point, on ROC-AUC and on log-loss alike, and cheaper selected points (supplementary Section S10). These are different points of the same trade-off rather than a contradiction of indicator results computed over whole fronts. They do not support the converse either: on these four datasets a better approximation of the front does not reliably buy a better selected solution, and the two contrasts where the pick does improve are the ones with the smaller indicator effects. Any statement that out-of-fold selection transfers must therefore be read as within-method and in level, not as evidence that better fronts give better picks.

7. Limitations

Four datasets is the binding constraint. The statistical unit is one outer partition, but the unit of *generalization* is the dataset, and there are four. Every replication-level interval here describes partition sensitivity within a dataset, not uncertainty about a population of datasets, so every cross-dataset mechanism statement is an association over four points and not a tested hypothesis; in particular, the benefit of real anchors does not track surrogate reliability (Section 5.3). All four datasets are binary tabular problems of 3×10^4 to 2×10^5 rows from financial-services applications, three of them from Kaggle competitions, so the four points are not four independent domains.

Nothing here establishes behaviour outside financial services, on multiclass problems, text or image data, very small samples or ensembles of more than five members, where the simplex and design grow quickly, and every conclusion is scoped to the one-shot, no-infill case (Section 2.5). Repeating the unchanged protocol on a broader, prospectively specified panel that includes other domains is a natural next step.

The partitions ran in two batches. Partitions 0–9 and 10–29 ran as two separate batches with the same recorded platform and core count, and some indicator levels and anchor win rates differ between them (Section 5.9). The recorded provenance does not include the machine's power or thermal state, its concurrent load or thread-related environment variables during either batch, so we report the difference as exploratory and cannot explain it; every interval and test here treats the 30 partitions of a dataset as exchangeable.

Neither NBI contrast is a single-factor experiment. Replacing the anchors changes four things at once (Section 3.4). The injection control of Section 5.3 separates the injected extreme points from the other three, but no control in this study separates the utopia point, the normalization and the direction set from one another; that would require an NBI implementation in which each can be varied independently. NBI-B versus NBI-C also changes the solver acceptance rule, because certified optimality is generically unattainable on a piecewise-constant ROC-AUC, so success rates are not on a common scale across arms, and the UCI credit gap between NBI-C and NBI-B is substantially a convergence artifact of NBI-B (Sections 5.4 and 6.4).

The model zoo is fixed and shapes the findings. Base learners and hyperparameters were fixed in advance with no search: the study measures the weighting layer with the model layer held constant, so every empirical result here is conditional on this zoo. The zoo contains a member that is poorly calibrated on three of the four datasets and an expensive one (k-nearest neighbours alone costs more than the other four combined in every replication; Section 4.2), which makes the cost objective non-trivial and also drives the β_{ij} result of Section 5.2: the largest interactions attach to the weakest vertex. Whether the surface, anchor, cost and coefficient results hold for a zoo of near-equal members is untested and left to future work.

The Pareto reference is empirical. With no analytical front, all convergence and coverage indicators are computed against a sample-based reference of about 10^5 points per replication. Its weighted-cost front met the 5% displacement tolerance in 118 of 120 replications and ended at most 6% displaced in the other two (supplementary Section S8); the convergence check does not cover the support-cost front. The final reference also contains every candidate set, so methods contributing many points are partly graded against their own output. Those points lower every arm's hypervolume ratio rather than inflating it, and can change a hypervolume contrast only through the normalization box (Section 3.7); under IGD^+ they can favour their own arm. Against the sample core no NBI-B versus NBI-A direction changes, but two small IGD^+ readings reverse: NBI-C versus NBI-B on BNP Paribas and NBI-C versus random scalarization on Santander (Section 5.4).

No comparison is compute-matched. The arms are compared as specified, not at equal budget, so the comparison supports no efficiency claim (Table 5). The only external baseline, NSGA-II (Section 5.5), is one canonical algorithm in one untuned configuration, and nothing here ranks it against MOEA/D, NSGA-III or a surrogate-assisted evolutionary method. Its budget was set from NBI-C's evaluation counter, which records the ROC-AUC and the log-loss computation of each weight vector separately while NSGA-II's counts weight vectors, so NSGA-II evaluated about twice as many weight vectors as NBI-C's solve; it also used about 2.2× the wall clock of that solve, with its runs executed eight at a time, and returned 66 non-dominated points against

NBI-C medians of 34 to 65, so its margins should be read as an upper bound on its advantage. It outperformed the metamodel-free arm on every dataset. Whether NSGA-II also leads at an equal number of weight-vector evaluations was not tested. The low-budget question, what an evolutionary method or a random-search ladder achieves on the 66 design points and 100 gate points the surrogate arms use, is the one a surrogate paper most needs and the one we do not answer.

The cost objective is a measured proxy. All c_i are wall-clock medians on one workstation with a fixed library stack, one thread configuration and a per-dataset batch size; both cost models ignore memory, model loading, shared preprocessing and parallel execution of the members, and the c_i would change with hardware, thread allocation, batching, a compiled inference runtime, or an approximate-nearest-neighbour index replacing brute-force search, which would remove the study's largest cost asymmetry. The support cost adds the threshold $\varepsilon = 10^{-3}$ of Equation (6), fixed before the study and never changed: an AUC-inert k-nearest-neighbour weight just above it produced a spurious difference of about 250 ms per 1,000 rows (Section 5.6), so any support cost needs a weight-cleaning step. Conclusions that depend on the support cost are sensitive to ε and to the cost range that sets the normalization box, and are conditional on the active members running one after another (Section 3.2). The cost objective is realistic for the setting measured, not universal.

No claim of universal dominance, and confounded comparators. Among the surrogate-derived methods, NBI-C is best or tied-best on the primary endpoints under the weighted cost on these four datasets; it ties with random scalarization on UCI credit, is not statistically separated from it on Santander IGD^+, where the direction depends on the reference (Section 5.4), and loses its support-cost advantage on BNP Paribas and UCI credit. Its advantage over scalarization on BNP Paribas and Porto Seguro is confounded with the objective source: NBI-C optimizes exact objectives, scalarization optimizes surrogates, and separating the two would require a scalarization arm on the cached out-of-fold objectives, which we did not run.

Selection rules and the holdout analysis. Holdout transfer is verified only within a candidate set and in level, for a knee rule under the weighted cost with TOPSIS and the support cost as sensitivity analyses (Section 5.8). Whether the arm with the better front also yields the better selected solution is answered only descriptively, and in the negative at the selected point on the two NBI contrasts that are Holm-significant on both primary endpoints (Section 6.8).

8. Conclusion

We transferred an established DoE–RSM–NBI framework from multiobjective engineering optimization and forecasting to the design of weighted classifier ensembles under a measured deployment cost, and asked when its Pareto front can be believed. Three NBI arms, defined by the source of the objectives and the source of the anchors, were compared over four public binary tabular datasets $\times$ 30 partitions, with every candidate revalidated on the true out-of-fold objectives and scored against an empirical reference. The main findings, each limited to one-shot pipelines without infill, one fixed zoo of five classifiers and four datasets from one application domain, are the following.

- Surrogate adequacy depends on the dataset and the metric and must be measured on unseen compositions. ROC-AUC surfaces passed the external reliability gate in 0, 5, 16 and 30 of 30 partitions on Santander, BNP Paribas, Porto Seguro and UCI credit, and log-loss surfaces in 30, 3, 12 and 30; smoothness of the response in the weights does not explain the difference on its own.

- On Santander and BNP Paribas, anchor misplacement had the largest observed effect among failures attributable to the surrogate pipeline. Under the continuous cost the arms optimize, real anchors improved both primary endpoints in all 30 partitions of these two datasets, in 23 to 24 of 30 on Porto Seguro and in 24 of 30 on UCI credit, although the improvement is significant after the Nadeau–Bengio correction and Holm adjustment only on Santander. On Santander and BNP Paribas most of the hypervolume gain comes from returning the real single-objective optima rather than from the relocated construction, and removing the metamodel once the anchors are right adds far less than the anchors there, about as much on Porto Seguro and three times as much on UCI credit, where a subproblem-convergence effect of the real-anchored surrogate arm (NBI-B) rather than surrogate error dominates; this ordering is an observation, not a tested claim.

- The gate screens surface accuracy, not anchor placement. On BNP Paribas, surrogate-anchored NBI (NBI-A) collapses in exactly the seven partitions where the parsimony rule selects a quadratic ROC-AUC surface, which it does when that surface fits unseen compositions better, and these partitions pass the ROC-AUC gate more often than the rest (3 of 7 against 2 of 23); the gate therefore needs an accompanying anchor check.

- The classical synergism criterion $\beta_{ij} > |\beta_i - \beta_j|$ holds for the leading classifier pair in every partition of all four datasets, yet the real 50/50 blend beats its better member on only one of them, so a large interaction coefficient must be confirmed on the real objective before it is read as complementarity.

- The continuous cost the methods optimize and the step cost deployment pays disagree about the hypervolume-best of six candidate sets in 8 to 24 of 30 partitions, and several paired conclusions, including the UCI credit anchor effect, reverse between them. Sets optimized on the linear relaxation omit the cheapest supports that the unoptimized design contains, at a penalty whose size depends on the cost range that sets the normalization box.

- On the same three real objectives, NSGA-II approximated the front better than metamodel-free NBI (NBI-C) in all eight dataset–endpoint comparisons, five of them significant after correction, and gave more evenly spaced revalidated fronts. It used about twice NBI-C's weight-vector evaluations and about 2.2 times its wall clock (NSGA-II runs executed eight at a time), and whether it also leads at an equal number of weight-vector evaluations was not tested. Its returned sets are larger, which mechanically favours it on both endpoints, so its small margins should be read as an upper bound on its advantage. This is one untuned configuration on four datasets from one domain, not a general recommendation; it bounds what the NBI pipeline is worth, not the diagnosis of when it can be trusted.

What we carry forward is therefore a set of controls rather than a method: validate the surface on compositions it has not seen; compare each surrogate-derived anchor with the optimum computed directly on the out-of-fold objective and make sure the latter reaches the returned set; revalidate every candidate on the out-of-fold objectives before scoring; build a reference whose sampled core contains no output of the compared methods; and state which cost is being paid. Each changed a conclusion that would otherwise have been reported, and the anchor check is not free (Table 5). The response surfaces remain a reproducible description of the ensemble-weight simplex: their four largest interaction coefficients keep their sign across all thirty resamplings, the coefficient of variation of the largest one lies between 0.007 and 0.17 across the eight dataset–metric surfaces, and they say something interpretable about which components are weak. Reproducibility is not accuracy, however, and their fitted values deserve trust only where a surface has been validated on compositions it has not seen; whether they are also a sound optimization substrate must be checked on each problem with these controls.

Replication mattered: the first ten partitions gave the direction of every primary effect that is statistically established at $R = 30$, but left the Porto Seguro and UCI credit anchor effect undecided (6/10 and 5/10 on hypervolume), which only thirty partitions showed to be consistent in direction.

Reproducibility statement

Code and frozen states. Code, manifests, aggregated tables, statistics and figures are in the repository [repository name withheld for anonymized review]. Two annotated tags freeze the study: `pco213-postwork-r10` the ten-replication analysis and `pco213-postwork-r30` (commit `b3ed050`) the thirty-replication analysis reported here (Section 5.9 compares the two tags). The manuscript, figures and tables are on the branch `paper/surrogate-nbi-ensemble-weighting`, cut from that tag, together with the results computed after it from the stored outputs of the thirty-replication run, among them the NSGA-II baseline, the anchor-injection control and the blend test of Table 3. A single checkpointed ten-stage runner drives the benchmark and resumes an interrupted run without recomputation; its stages, the resumption planner, the aggregation and statistics scripts and the coverage of the library's 83 unit tests are in supplementary Section S1. The commit history records that the library, the benchmark runner and the analysis and asset-building scripts were developed with the assistance of Anthropic Claude models (Claude Fable 5, Claude Fable 5.1 and Claude Opus 5) used through Claude Code; the library's behaviour is checked by those unit tests.

Determinism and environment. Every stochastic component is seeded as $20260904 + r$ for replication r; the offsets, the seed map, the configuration history and the resolved library versions are in the master manifest (supplementary Section S1). The environment is Python 3.11.15 with NumPy 2.4.6 (Harris et al., 2020), pandas 3.0.5, SciPy 1.17.1 (Virtanen et al., 2020), scikit-learn 1.9.0 (Pedregosa et al., 2011), statsmodels 0.15.0 and XGBoost 3.2.0 (Chen and Guestrin, 2016), on macOS (Apple M4 Max, 14 cores, arm64). Base-model inference is timed under the thread setting of Section 4.2, the batched weight evaluations of the empirical-reference stage use eight worker threads, and the NSGA-II runs, except for one pilot run, were executed eight at a time as concurrent processes. Execution provenance (stage time and the fit, design, subproblem and objective-evaluation counts) is in supplementary Section S1. Two stages were retried, both in replication 0 after the solver fix described there; none in the extension.

Data availability and redistribution limits. The raw datasets are not redistributed: the UCI default-of-credit-card-clients data are downloadable from the UCI Machine Learning Repository, while the three Kaggle datasets are subject to competition rules and must be obtained from Kaggle by the user. Per-dataset manifests record the provenance (source, acquisition command, expected invariants and the SHA-256 checksum of the file actually used), so a copy of the same file can be verified byte-for-byte (supplementary Section S1). All three Kaggle training files were obtained from public mirror datasets hosted on Kaggle and validated against those invariants; the manifests name each mirror. Competition test labels are never used. Porto Seguro was subsampled to 200,000 rows by a stratified draw with the recorded seed. The per-replication raw artifacts total about 829 MB and are deliberately not version-controlled, but are reconstructible from the seeds and the runner, except for the wall-clock costs c_i, which are re-measured on each run, and the results that depend on them; the measured costs are committed in `tables/inference_costs.csv`.

What a replication requires. Reproducing the aggregate results from the published tables and statistics requires only the repository; from raw data it requires the four datasets, the pinned environment and approximately 81 hours of single-workstation compute for the benchmark plus

about 15.5 hours, with eight concurrent processes, for the NSGA-II baseline, or proportionally less for a subset of datasets or replications, since the benchmark runner is resumable and parameterized by dataset and replication count.

Funding and acknowledgements are given on the separate title page.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Anthropic Claude (Claude Opus 5, through Claude Code) and OpenAI ChatGPT in order to assist with drafting, restructuring and editing the text, language refinement, consistency checking of the text against the stored results, collation and checking of the references, and the AI-run pre-submission review of the manuscript and its verification. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. The use of AI assistance in writing the analysis code is stated in the Reproducibility statement.

References

Abbass, H.A., 2003. Pareto neuro-evolution: Constructing ensemble of neural networks using multi-objective optimization, in: The 2003 Congress on Evolutionary Computation, 2003. CEC '03, pp. 2074–2080. doi:10.1109/CEC.2003.1299928.

Abolghasemian, M., Ghane Kanafi, A., Daneshmand-Mehr, M., 2022. Simulation-based multi-objective optimization of open-pit mine haulage system: A modified-NBI method and meta modeling approach. Complexity 2022, 3540736. doi:10.1155/2022/3540736.

de Almeida, F.A., Santos, A.C.O., de Paiva, A.P., Gomes, G.F., de Freitas Gomes, J.H., 2022. Multivariate Taguchi loss function optimization based on principal components analysis and normal boundary intersection. Engineering with Computers 38, 1627–1643. doi:10.1007/s00366-020-01122-8.

Bacci, L.A., Mello, L.G., Incerti, T., de Paiva, A.P., Balestrassi, P.P., 2019. Optimization of combined time series methods to forecast the demand for coffee in Brazil: A new approach using Normal Boundary Intersection coupled with mixture designs of experiments and rotated factor scores. International Journal of Production Economics 212, 186–211. doi:10.1016/j.ijpe.2019.03.001.

Bandyopadhyay, S., Saha, S., Maulik, U., Deb, K., 2008. A simulated annealing-based multiobjective optimization algorithm: AMOSA. IEEE Transactions on Evolutionary Computation 12, 269–283. doi:10.1109/TEVC.2007.900837.

Belinato, G., de Almeida, F.A., de Paiva, A.P., de Freitas Gomes, J.H., Balestrassi, P.P., Rosa, P.A.R.C., 2019. A multivariate normal boundary intersection PCA-based approach to reduce dimensionality in optimization problems for LBM process. Engineering with Computers 35, 1533–1544. doi:10.1007/s00366-018-0678-3.

Blank, J., Deb, K., 2020. Pymoo: Multi-objective optimization in Python. IEEE Access 8, 89497–89509. doi:10.1109/ACCESS.2020.2990567.

Borchert, O., Salinas, D., Flunkert, V., Januschowski, T., Günnemann, S., 2022. Multi-objective model selection for time series forecasting. arXiv preprint arXiv:2202.08485. 10.48550/arXiv.2202.08485.

Breiman, L., 1996. Stacked regressions. Machine Learning 24, 49–64. doi:10.1007/BF00117832.

Brito, T.G., Paiva, A.P., Ferreira, J.R., Gomes, J.H.F., Balestrassi, P.P., 2014. A normal boundary intersection approach to multiresponse robust optimization of the surface roughness in end milling process with combined arrays. Precision Engineering 38, 628–638. doi:10.1016/j.precisioneng.2014.02.013.

Brown, L.D., Cai, T.T., DasGupta, A., 2001. Interval estimation for a binomial proportion. Statistical Science 16, 101–133. doi:10.1214/ss/1009213286.

Caruana, R., Niculescu-Mizil, A., Crew, G., Ksikes, A., 2004. Ensemble selection from libraries of models, in: Proceedings of the Twenty-First International Conference on Machine Learning (ICML '04), ACM. p. 18. doi:10.1145/1015330.1015432.

Chandra, A., Yao, X., 2006. Ensemble learning using multi-objective evolutionary algorithms. Journal of Mathematical Modelling and Algorithms 5, 417–445. doi:10.1007/s10852-005-9020-3.

Chen, L., Zaharia, M., Zou, J., 2020. FrugalML: How to use ML prediction APIs more accurately and cheaply, in: Advances in Neural Information Processing Systems 33 (NeurIPS 2020). ArXiv:2006.07512.

Chen, M.F., Hu, M.Y., Lourie, N., Cho, K., Ré, C., 2025. Aioli: A Unified Optimization Framework for Language Model Data Mixing, in: The Thirteenth International Conference on Learning Representations (ICLR 2025). ArXiv:2411.05735.

Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system, in: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16), ACM. pp. 785–794. doi:10.1145/2939672.2939785.

Chugh, T., Jin, Y., Miettinen, K., Hakanen, J., Sindhya, K., 2018. A surrogate-assisted reference vector guided evolutionary algorithm for computationally expensive many-objective optimization. IEEE Transactions on Evolutionary Computation 22, 129–142. doi:10.1109/TEVC.2016.2622301.

Chugh, T., Sindhya, K., Hakanen, J., Miettinen, K., 2019. A survey on handling computationally expensive multiobjective optimization problems with evolutionary algorithms. Soft Computing 23, 3137–3166. doi:10.1007/s00500-017-2965-0.

Cornell, J.A., 2002. Experiments with Mixtures: Designs, Models, and the Analysis of Mixture Data. Wiley Series in Probability and Statistics, Wiley, Hoboken, NJ. doi:10.1002/9781118204221.

Costa, D.M.D., Brito, T.G., de Paiva, A.P., Leme, R.C., Balestrassi, P.P., 2016. A normal boundary intersection with multivariate mean square error approach for dry end milling process optimization of the AISI 1045 steel. Journal of Cleaner Production 135, 1658–1672. doi:10.1016/j.jclepro.2016.01.062.

Das, I., Dennis, J.E., 1998. Normal-Boundary Intersection: A new method for generating the Pareto surface in nonlinear multicriteria optimization problems. SIAM Journal on Optimization 8, 631–657. doi:10.1137/S1052623496307510.

Deb, K., Hussein, R., Roy, P.C., Toscano-Pulido, G., 2019. A taxonomy for metamodeling frameworks for evolutionary multiobjective optimization. IEEE Transactions on Evolutionary Computation 23, 104–116. doi:10.1109/TEVC.2018.2828091.

Deb, K., Miettinen, K., Chaudhuri, S., 2010. Toward an estimation of nadir objective vector using a hybrid of evolutionary and local search approaches. IEEE Transactions on Evolutionary Computation 14, 821–841. doi:10.1109/TEVC.2010.2041667.

Deb, K., Pratap, A., Agarwal, S., Meyarivan, T., 2002. A fast and elitist multiobjective genetic algorithm: NSGA-II. IEEE Transactions on Evolutionary Computation 6, 182–197. doi:10.1109/4235.996017.

Demšar, J., 2006. Statistical comparisons of classifiers over multiple data sets. Journal of Machine Learning Research 7, 1–30.

Dietterich, T.G., 2000. Ensemble methods in machine learning, in: Multiple Classifier Systems (MCS 2000), Lecture Notes in Computer Science, Springer. pp. 1–15. doi:10.1007/3-540-45014-9_1.

Dos Santos, E.M., Sabourin, R., Maupin, P., 2008. A dynamic overproduce-and-choose strategy for the selection of classifier ensembles. Pattern Recognition 41, 2993–3009. doi:10.1016/j.patcog.2008.03.027.

Efron, B., Tibshirani, R.J., 1993. An Introduction to the Bootstrap. Chapman & Hall/Springer, New York. doi:10.1007/978-1-4899-4541-9.

Ekbal, A., Saha, S., 2011. A multiobjective simulated annealing approach for classifier ensemble: Named entity recognition in Indian languages as case studies. Expert Systems with Applications 38, 14760–14772. doi:10.1016/j.eswa.2011.05.004.

Ekbal, A., Saha, S., 2013. Combining feature selection and classifier ensemble using a multiobjective simulated annealing approach: Application to named entity recognition. Soft Computing 17, 1–16. doi:10.1007/s00500-012-0885-6.

Erickson, N., Mueller, J., Shirkov, A., Zhang, H., Larroy, P., Li, M., Smola, A., 2020. AutoGluon-Tabular: Robust and accurate AutoML for structured data. arXiv preprint arXiv:2003.06505. 10.48550/arXiv.2003.06505.

Feurer, M., Eggensperger, K., Falkner, S., Lindauer, M., Hutter, F., 2022. Auto-Sklearn 2.0: Hands-free AutoML via meta-learning. Journal of Machine Learning Research 23. ArXiv:2007.04074.

Fumera, G., Roli, F., 2005. A theoretical and experimental analysis of linear combiners for multiple classifier systems. IEEE Transactions on Pattern Analysis and Machine Intelligence 27, 942–956. doi:10.1109/TPAMI.2005.109.

Galván, J., Sánchez, G., Jiménez, F., 2026. Simultaneous multi-objective evolutionary optimization of heterogeneous ensembles, learner-specific feature subsets, and aggregation weights. Algorithms 19, 681. doi:10.3390/a19080681.

Gellerich, P.A., Majschak, J.P., 2023. A design of experiments based on the Normal-boundary-intersection method to identify optimum machine settings in manufacturing processes. PAMM 23, e202300009. doi:10.1002/pamm.202300009.

Gunasekaran, J.R., Mishra, C.S., Thinakaran, P., Sharma, B., Kandemir, M.T., Das, C.R., 2022. Cocktail: A multidimensional optimization for model serving in cloud, in: 19th USENIX Symposium on Networked Systems Design and Implementation (NSDI 22), USENIX Association, Renton, WA. pp. 1041–1057.

Hansen, L.K., Salamon, P., 1990. Neural network ensembles. IEEE Transactions on Pattern Analysis and Machine Intelligence 12, 993–1001. doi:10.1109/34.58871.

Harris, C.R., Millman, K.J., van der Walt, S.J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N.J., Kern, R., Picus, M., Hoyer, S., van

Kerkwijk, M.H., Brett, M., Haldane, A., Fernández del Río, J., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C., Oliphant, T.E., 2020. Array programming with NumPy. Nature 585, 357–362. doi:10.1038/s41586-020-2649-2.

Hassan, A., Mohamed, R., Abdel-Basset, M., Abouhawwash, M., 2026. A weighted voting-based ensemble classifier based on manta ray foraging optimizer for cyberattack detection in IoT environments: A comparative study. Evolutionary Intelligence 19, 18. doi:10.1007/s12065-025-01126-7.

He, L., Ishibuchi, H., Trivedi, A., Wang, H., Nan, Y., Srinivasan, D., 2021. A survey of normalization methods in multiobjective evolutionary algorithms. IEEE Transactions on Evolutionary Computation 25, 1028–1048. doi:10.1109/TEVC.2021.3076514.

Herrmann-Wicklmayr, M., Flaßkamp, K., 2026. Non-extreme individual minima for improved Pareto front sampling efficiency and decision-making, in: Computational Science – ICCS 2026 Workshops, Lecture Notes in Computer Science. Springer Nature Switzerland, pp. 271–284. doi:10.1007/978-3-032-29909-3_20. arXiv:2602.21883.

Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6, 65–70.

Isermann, H., Steuer, R.E., 1988. Computational experience concerning payoff tables and minimum criterion values over the efficient set. European Journal of Operational Research 33, 91–97. doi:10.1016/0377-2217(88)90257-3.

Ishibuchi, H., Masuda, H., Tanigaki, Y., Nojima, Y., 2015. Modified distance calculation in generational distance and inverted generational distance, in: Evolutionary Multi-Criterion Optimization (EMO 2015), Lecture Notes in Computer Science, Springer. pp. 110–125. doi:10.1007/978-3-319-15892-1_8.

Ji, J., Li, J., 2023. Tri-objective optimization-based cascade ensemble pruning for deep forest. Pattern Recognition 143, 109744. doi:10.1016/j.patcog.2023.109744.

Jin, Y., 2011. Surrogate-assisted evolutionary computation: Recent advances and future challenges. Swarm and Evolutionary Computation 1, 61–70. doi:10.1016/j.swevo.2011.05.001.

Jin, Y., Sendhoff, B., 2008. Pareto-based multiobjective machine learning: An overview and case studies. IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews) 38, 397–415. doi:10.1109/TSMCC.2008.919172.

Kaggle, 2016. BNP Paribas Cardif claims management. Competition page, https://www.kaggle.com/competitions/bnp-paribas-cardif-claims-management. Accessed 9 September 2026.

Kaggle, 2017. Porto Seguro's safe driver prediction. Competition page, https://www.kaggle.com/competitions/porto-seguro-safe-driver-prediction. Accessed 9 September 2026.

Kaggle, 2019. Santander customer transaction prediction. Competition page, https://www.kaggle.com/competitions/santander-customer-transaction-prediction. Accessed 9 September 2026.

Karl, A.T., Essex, S., Wisnowski, J., Rushing, H., 2023. A Workflow for Lipid Nanoparticle (LNP) Formulation Optimization using Designed Mixture-Process Experiments and Self-Validated Ensemble Models (SVEM). Journal of Visualized Experiments doi:10.3791/65200. also available as arXiv:2212.11264.

Kerby, D.S., 2014. The simple difference formula: An approach to teaching nonparametric correlation. Comprehensive Psychology 3, 11.IT.3.1. doi:10.2466/11.IT.3.1.

Kittler, J., Hatef, M., Duin, R.P.W., Matas, J., 1998. On combining classifiers. IEEE Transactions on Pattern Analysis and Machine Intelligence 20, 226–239. doi:10.1109/34.667881.

Knowles, J., 2006. ParEGO: A hybrid algorithm with on-line landscape approximation for expensive multiobjective optimization problems. IEEE Transactions on Evolutionary Computation 10, 50–66. doi:10.1109/TEVC.2005.851274.

Kraft, D., 1988. A Software Package for Sequential Quadratic Programming. Technical Report DFVLR-FB 88-28. DLR German Aerospace Center.

Kuncheva, L.I., 2014. Combining Pattern Classifiers: Methods and Algorithms. 2nd ed., Wiley, Hoboken, NJ. doi:10.1002/9781118914564.

Kwon, Y., Lee, K., Lee, D., 2024. Ensemble Prediction Model Using Mixture Design of Experiments. Journal of Society of Korea Industrial and Systems Engineering 47, 161–170. doi:10.11627/jksie.2024.47.4.161.

van der Laan, M.J., Polley, E.C., Hubbard, A.E., 2007. Super learner. Statistical Applications in Genetics and Molecular Biology 6, Article 25. doi:10.2202/1544-6115.1309.

Leal, G.d.S., Romão, E.L., Reis, D.L.d.P.E.d., Balestrassi, P.P., de Paiva, A.P., 2022. A multiobjective portfolio optimization for energy assets using D-Optimal design and mixture design of experiments. Production 32, e20210119. doi:10.1590/0103-6513.20210119.

Leal, G.S., Bessegato, L.F., Xavier, Y.S.M., Melgani, F., Balestrassi, P.P., 2025. A design-of-experiments-based approach for efficient estimation of bimodal Gaussian mixture weights. IEEE Access 13, 168322–168334. doi:10.1109/ACCESS.2025.3614023.

LeBlanc, M., Tibshirani, R., 1996. Combining estimates in regression and classification. Journal of the American Statistical Association 91, 1641–1650. doi:10.1080/01621459.1996.10476733.

LeDell, E., van der Laan, M.J., Petersen, M., 2016. AUC-maximizing ensembles through metalearning. The International Journal of Biostatistics 12, 203–218. doi:10.1515/ijb-2015-0035.

Liu, Q., Zheng, X., Muennighoff, N., Zeng, G., Dou, L., Pang, T., Jiang, J., Lin, M., 2025. RegMix: Data Mixture as Regression for Language Model Pre-training, in: The Thirteenth International Conference on Learning Representations (ICLR 2025). ArXiv:2407.01492.

Lopes, L.G.D., Brito, T.G., Paiva, A.P., Peruchi, R.S., Balestrassi, P.P., 2016. Robust parameter optimization based on multivariate normal boundary intersection. Computers & Industrial Engineering 93, 55–66. doi:10.1016/j.cie.2015.12.023.

Maier, J., Möller, F., Purucker, L., 2024. Hardware aware ensemble selection for balancing predictive accuracy and cost, in: Third International Conference on Automated Machine Learning (AutoML 2024), Workshop Track. ArXiv:2408.02280.

Maier, J., Purucker, L., 2026. HAPEns: Hardware-aware post-hoc ensembling for tabular data. arXiv preprint arXiv:2603.10582v2. 10.48550/arXiv.2603.10582.

Mendes, R.R.A., Paiva, A.P., Peruchi, R.S., Balestrassi, P.P., Leme, R.C., Silva, M.B., 2016. Multiobjective portfolio optimization of ARMA–GARCH time series based on experimental designs. Computers & Operations Research 66, 434–444. doi:10.1016/j.cor.2015.05.001.

Messac, A., Ismail-Yahaya, A., Mattson, C.A., 2003. The normalized normal constraint method for generating the Pareto frontier. Structural and Multidisciplinary Optimization 25, 86–98. doi:10.1007/s00158-002-0276-1.

Moreira, L.F.R., Moreira, R., Rodrigues, L.G.F., 2026. Energy-aware ensemble learning for coffee leaf disease classification, in: 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC), pp. 1–6. doi:10.1109/CCNC65079.2026.11366365. arXiv:2601.12109.

Moreira, M.O., Balestrassi, P.P., Paiva, A.P., Ribeiro, P.F., Bonatto, B.D., 2021. Design of experiments using artificial neural network ensemble for photovoltaic generation forecasting. Renewable and Sustainable Energy Reviews 135, 110450. doi:10.1016/j.rser.2020.110450.

Myers, R.H., Montgomery, D.C., Anderson-Cook, C.M., 2016. Response Surface Methodology: Process and Product Optimization Using Designed Experiments. Wiley Series in Probability and Statistics. 4th ed., John Wiley & Sons, Hoboken, NJ.

Nadeau, C., Bengio, Y., 2003. Inference for the generalization error. Machine Learning 52, 239–281. doi:10.1023/A:1024068626366.

Nan, F., Saligrama, V., 2017. Adaptive classification for prediction under a budget, in: Advances in Neural Information Processing Systems 30 (NIPS 2017), pp. 4727–4737. ArXiv:1705.10194.

Nan, F., Wang, J., Saligrama, V., 2016. Pruning random forests for prediction on a budget, in: Advances in Neural Information Processing Systems 29 (NIPS 2016), Curran Associates, Inc. URL: https://proceedings.neurips.cc/paper/2016/hash/3948ead63a9f2944218de038d8934305-Abstract.html. arXiv:1606.05060.

Naves, F.L., de Paula, T.I., Balestrassi, P.P., Braga, W.L.M., Sawhney, R.S., de Paiva, A.P., 2017. Multivariate Normal Boundary Intersection based on rotated factor scores: A multiobjective optimization method for methyl orange treatment. Journal of Cleaner Production 143, 413–439. doi:10.1016/j.jclepro.2016.12.092.

Onan, A., Korukoğlu, S., Bulut, H., 2016. A multiobjective weighted voting ensemble classifier based on differential evolution algorithm for text sentiment classification. Expert Systems with Applications 62, 1–16. doi:10.1016/j.eswa.2016.06.005.

de Paula, T.I., Gomes, G.F., de Freitas Gomes, J.H., de Paiva, A.P., 2019. A mixture design of experiments approach for genetic algorithm tuning applied to multi-objective optimization, in: Optimization of Complex Systems: Theory, Models, Algorithms and Applications. Springer International Publishing. Advances in Intelligent Systems and Computing, pp. 600–610. doi:10.1007/978-3-030-21803-4_60.

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., Duchesnay, É., 2011. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

Pereira, M.C., Tertuliano Ribeiro, C., Mendes, R.R.A., Campos, P.H.d.S., de Paiva, A.P., 2025. A hybrid multivariate normal boundary intersection approach with post-optimization assisted by mixture design of experiments. Engineering Applications of Artificial Intelligence 162, 112510. doi:10.1016/j.engappai.2025.112510.

Piepel, G.F., 1982. Measuring Component Effects in Constrained Mixture Experiments. Technometrics 24, 29–39. doi:10.1080/00401706.1982.10487706.

Ponweiser, W., Wagner, T., Biermann, D., Vincze, M., 2008. Multiobjective optimization on a limited budget of evaluations using model-assisted S-metric selection, in: Parallel Problem Solving from Nature – PPSN X, Lecture Notes in Computer Science. Springer Berlin Heidelberg, pp. 784–794. doi:10.1007/978-3-540-87700-4_78.

Qian, C., Yu, Y., Zhou, Z.H., 2015. Pareto ensemble pruning. Proceedings of the AAAI Conference on Artificial Intelligence 29. doi:10.1609/aaai.v29i1.9579.

Rezaei, K., Agahi, H., Mahmoodzadeh, A., 2019. Multi-objective differential evolution-based ensemble method for brain tumour diagnosis. IET Image Processing 13, 1421–1430. doi:10.1049/iet-ipr.2018.6377.

Ribeiro, C.T., 2026. Meta-otimização dos hiperparâmetros do algoritmo XGBoost para classificação binária: uma integração entre planejamento de experimentos e o método da interseção normal à fronteira. Master's thesis. Universidade Federal de Itajubá, Programa de Pós-Graduação em Engenharia de Produção. Itajubá, Brazil. Advisor: Anderson Paulo de Paiva. 153 f. Available at https://repositorio.unifei.edu.br/jspui/handle/123456789/4372.

Ribeiro, V.H.A., Reynoso-Meza, G., 2020. Ensemble learning by means of a multi-objective optimization design approach for dealing with imbalanced data sets. Expert Systems with Applications 147, 113232. doi:10.1016/j.eswa.2020.113232.

Rocha, F.B., Rotella, P., Balestrassi, P.P., Melgani, F., Zambroni De Souza, A.C., 2025. Optimizing neural network ensemble for short-term load forecasting: A case study of a Brazilian electrical distribution system. IEEE Access 13, 207903–207915. doi:10.1109/ACCESS.2025.3641712.

Scheffé, H., 1958. Experiments with mixtures. Journal of the Royal Statistical Society: Series B (Statistical Methodology) 20, 344–360. doi:10.1111/j.2517-6161.1958.tb00299.x.

Scheffé, H., 1963. The simplex-centroid design for experiments with mixtures. Journal of the Royal Statistical Society: Series B (Statistical Methodology) 25, 235–251. doi:10.1111/j.2517-6161.1963.tb00506.x.

Schott, J.R., 1995. Fault Tolerant Design Using Single and Multicriteria Genetic Algorithm Optimization. Master's thesis. Massachusetts Institute of Technology, Department of Aeronautics and Astronautics. Thesis (M.S.); DSpace@MIT handle http://hdl.handle.net/1721.1/11582.

Shahhosseini, M., Hu, G., Pham, H., 2022. Optimizing ensemble weights and hyperparameters of machine learning models for regression problems. Machine Learning with Applications 7, 100251. doi:10.1016/j.mlwa.2022.100251.

Tabatabaei, M., Hakanen, J., Hartikainen, M., Miettinen, K., Sindhya, K., 2015. A survey on handling computationally expensive multiobjective optimization problems using surrogates: Non-nature inspired methods. Structural and Multidisciplinary Optimization 52, 1–25. doi:10.1007/s00158-015-1226-z.

Thudi, A., Rovers, E., Ruan, Y., Thrush, T., Maddison, C.J., 2025. MixMin: Finding Data Mixtures via Convex Minimization, in: Proceedings of the 42nd International Conference on Machine Learning (ICML 2025). ArXiv:2502.10510.

Trapeznikov, K., Saligrama, V., Castañón, D., 2013. Multi-stage classifier design. Machine Learning 92, 479–502. doi:10.1007/s10994-013-5349-4.

Turney, P.D., 2000. Types of cost in inductive concept learning, in: Workshop on Cost-Sensitive Learning at the Seventeenth International Conference on Machine Learning (ICML 2000), Stanford University, California, pp. 15–21. ArXiv:cs/0212034.

Van Veldhuizen, D.A., Lamont, G.B., 2000. Multiobjective evolutionary algorithms: Analyzing the state-of-the-art. Evolutionary Computation 8, 125–147. doi:10.1162/106365600568158.

Virtanen, P., Gommers, R., Oliphant, T.E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S.J., Brett, M., Wilson, J., Millman, K.J., Mayorov, N., Nelson, A.R.J., Jones, E., Kern, R., Larson, E., Carey, C.J., Polat, İ., Feng, Y., Moore, E.W., VanderPlas, J., Laxalde, D., Perktold, J., Cimrman, R., Henriksen, I., Quintero, E.A., Harris, C.R., Archibald, A.M., Ribeiro, A.H., Pedregosa, F., van Mulbregt, P., SciPy 1.0 Contributors, 2020. SciPy 1.0: Fundamental algorithms for scientific computing in Python. Nature Methods 17, 261–272. doi:10.1038/s41592-019-0686-2.

Vrugt, J.A., Clark, M.P., Diks, C.G.H., Duan, Q., Robinson, B.A., 2006. Multi-objective calibration of forecast ensembles using Bayesian model averaging. Geophysical Research Letters 33, 2006GL027126. doi:10.1029/2006GL027126.

Wang, X., Kondratyuk, D., Christiansen, E., Kitani, K.M., Alon, Y., Eban, E., 2022. Wisdom of committees: An overlooked approach to faster and more accurate models, in: International Conference on Learning Representations (ICLR 2022). ArXiv:2012.01988.

Wilcoxon, F., 1945. Individual comparisons by ranking methods. Biometrics Bulletin 1, 80–83. doi:10.2307/3001968.

Wilson, E.B., 1927. Probable inference, the law of succession, and statistical inference. Journal of the American Statistical Association 22, 209–212. doi:10.1080/01621459.1927.10502953.

Wolpert, D.H., 1992. Stacked generalization. Neural Networks 5, 241–259. doi:10.1016/S0893-6080(05)80023-1.

Xie, S.M., Pham, H., Dong, X., Du, N., Liu, H., Lu, Y., Liang, P., Le, Q.V., Ma, T., Yu, A.W., 2023. DoReMi: Optimizing Data Mixtures Speeds Up Language Model Pretraining, in: Advances in Neural Information Processing Systems 36 (NeurIPS 2023), pp. 69798–69818. doi:10.52202/075280-3059. arXiv:2305.10429.

Xu, B., Liu, W., Ding, K., Lu, Y., Cui, B., 2026. PSEO: Optimizing post-hoc stacking ensemble through hyperparameter tuning. Proceedings of the AAAI Conference on Artificial Intelligence 40, 27188–27196. doi:10.1609/aaai.v40i32.39934.

Xu, Z., Kusner, M.J., Weinberger, K.Q., Chen, M., 2013. Cost-sensitive tree of classifiers, in: Proceedings of the 30th International Conference on Machine Learning (ICML 2013), pp. 133–141. ArXiv:1210.2771.

Xu, Z., Weinberger, K., Chapelle, O., 2012. The greedy miser: Learning under test-time budgets, in: Proceedings of the 29th International Conference on Machine Learning (ICML 2012). ArXiv:1206.6451.

Yao, Y., Vehtari, A., Simpson, D., Gelman, A., 2018. Using stacking to average Bayesian predictive distributions (with discussion). Bayesian Analysis 13, 917–1007. doi:10.1214/17-BA1091.

Ye, J., Liu, P., Sun, T., Zhan, J., Zhou, Y., Qiu, X., 2025. Data Mixing Laws: Optimizing Data Mixtures by Predicting Language Modeling Performance, in: The Thirteenth International Conference on Learning Representations (ICLR 2025). ArXiv:2403.16952.

Yeh, I.C., 2009. Default of credit card clients [Dataset]. UCI Machine Learning Repository. 10.24432/C55S3H. CC BY 4.0.

Zhang, L., Zhou, W.D., 2011. Sparse ensembles using weighted combination methods based on linear programming. Pattern Recognition 44, 97–106. doi:10.1016/j.patcog.2010.07.021.

Zhang, Q., Liu, W., Tsang, E., Virginas, B., 2010. Expensive multiobjective optimization by MOEA/D with Gaussian process model. IEEE Transactions on Evolutionary Computation 14, 456–474. doi:10.1109/TEVC.2009.2033671.

Zhao, J., Jiao, L., Xia, S., Basto Fernandes, V., Yevseyeva, I., Zhou, Y., Emmerich, M.T.M., 2018. Multiobjective sparse ensemble learning by means of evolutionary algorithms. Decision Support Systems 111, 86–100. doi:10.1016/j.dss.2018.05.003.

Zhou, Z.H., Wu, J., Tang, W., 2002. Ensembling neural networks: Many could be better than all. Artificial Intelligence 137, 239–263. doi:10.1016/S0004-3702(02)00190-X.

Zitzler, E., Thiele, L., 1999. Multiobjective evolutionary algorithms: A comparative case study and the strength Pareto approach. IEEE Transactions on Evolutionary Computation 3, 257–271. doi:10.1109/4235.797969.

Zitzler, E., Thiele, L., Laumanns, M., Fonseca, C.M., Grunert da Fonseca, V., 2003. Performance assessment of multiobjective optimizers: An analysis and review. IEEE Transactions on Evolutionary Computation 7, 117–132. doi:10.1109/TEVC.2003.810758.